

mp4

April 18, 2025

```
[1]: # # you will be prompted with a window asking to grant permissions
# from google.colab import drive
# drive.mount("/content/drive")

[2]: # # fill in the path in your Google Drive in the string below. Note: do not
#     ↳ escape slashes or spaces
# import os
# datadir = "/content/assignment4"
# if not os.path.exists(datadir):
#     !ln -s "/content/drive/MyDrive/MP4/" $datadir # TODO: Fill your Assignment
#     ↳ 4 path
# os.chdir(datadir)
# !pwd

[ ]: # NOTE: Loading data from Google Drive is VERY SLOW. Therefore, we download the
#     ↳ data to a local storage
# specified by VOC_PATH.
# This means the data would have be re-downloaded everytime.
# Change this path if you are running this script locally
VOC_PATH="data"

# !chmod u+x ./download_data.sh
# !sed -i 's/\r//g' ./download_data.sh
# !cat ./download_data.sh
# !bash ./download_data.sh $VOC_PATH

[4]: import os
import random

import cv2
import numpy as np

import torch
from torch.utils.data import DataLoader

from src.resnet_yolo import resnet152
import yolo_loss
```

```

from src.dataset import VocDetectorDataset
from src.eval_voc import evaluate
from src.predict import predict_image
from src.config import VOC_CLASSES, COLORS
from kaggle_submission import output_submission_csv

import matplotlib.pyplot as plt
import collections

import time

%matplotlib inline
%load_ext autoreload
%autoreload 2

```

0.1 Initialization

```

[5]: device = torch.device("cuda:0" if torch.cuda.is_available() else "mps" if torch.
    ↪mps.is_available() else "cpu")
device

```

```

[5]: device(type='mps')

```

```

[6]: # YOLO network hyperparameters
B = 2 # number of bounding box predictions per cell
S = 14 # width/height of network output grid (larger than 7x7 from paper since ↪
    ↪we use a different network)

```

0.2 Start from here if you modified yolo_loss.py and wish to retrain

```

[7]: # We recommend tuning only the batch size; the remaining hyperparameters should ↪
    ↪work well
# with the default values. However, you are free to experiment with them.

learning_rate = 0.001
num_epochs = 50
batch_size = 12

# Yolo loss component coefficients (as given in Yolo v1 paper)
lambda_coord = 5
lambda_noobj = 0.5

```

0.3 Reading Pascal Data

Since Pascal is a small dataset (5000 in train+val) we have combined the train and val splits to train our detector. This is not typically a good practice, but we will make an exception in this case to be able to get reasonable detection results with a comparatively small object detection dataset.

The train dataset loader also using a variety of data augmentation techniques including random shift, scaling, crop, and flips. Data augmentation is slightly more complicated for detection datasets since the bounding box annotations must be kept consistent throughout the transformations.

Since the output of the detector network we train is an $S \times S \times (B \times 5 + C)$, we use an encoder to convert the original bounding box coordinates into relative grid bounding box coordinates corresponding to the expected output. We also use a decoder which allows us to convert the opposite direction into image coordinate bounding boxes.

```
[8]: file_root_train = os.path.join(VOC_PATH, 'VOCdevkit_2007/VOC2007/JPEGImages/')
      annotation_file_train = 'data/voc2007.txt'

      train_dataset = □
        ↪VocDetectorDataset(root_img_dir=file_root_train,dataset_file=annotation_file_train,train=True,
        ↪S=S)
      train_loader = □
        ↪DataLoader(train_dataset,batch_size=batch_size,shuffle=True,num_workers=2)
      print('Loaded %d train images' % len(train_dataset))
```

Initializing dataset

Loaded 5011 train images

```
[9]: file_root_test = os.path.join(VOC_PATH, 'VOCdevkit_2007/VOC2007test/JPEGImages/'
      ↪')
      annotation_file_test = 'data/voc2007test.txt'

      test_dataset = □
        ↪VocDetectorDataset(root_img_dir=file_root_test,dataset_file=annotation_file_test,train=False,
        ↪S=S)
      test_loader = □
        ↪DataLoader(test_dataset,batch_size=batch_size,shuffle=False,num_workers=2)
      print('Loaded %d test images' % len(test_dataset))
```

Initializing dataset

Loaded 4950 test images

```
[10]: data = train_dataset[0]
```

0.4 Initializing the network

To implement Yolo we will rely on a pretrained classifier as the backbone for our detection network. PyTorch offers a variety of models which are pretrained on ImageNet in the `torchvision.models` package. In particular, we will use the ResNet50 architecture as a base for our detector. This is different from the base architecture in the Yolo paper and also results in a different output grid size (14x14 instead of 7x7).

Models are typically pretrained on ImageNet since the dataset is very large (> 1 million images) and widely used. The pretrained model provides a very useful weight initialization for our detector, so that the network is able to learn quickly and effectively.

```
[11]: load_network_path = None
pretrained = True

# use to load a previously trained network
if load_network_path is not None:
    print('Loading saved network from {}'.format(load_network_path))
    net = resnet152().to(device)
    net.load_state_dict(torch.load(load_network_path))
else:
    print('Load pre-trained model')
    net = resnet152(pretrained=pretrained).to(device)
```

Load pre-trained model

Downloading: "https://download.pytorch.org/models/resnet152-f82ba261.pth" to
 /Users/edu/.cache/torch/hub/checkpoints/resnet152-f82ba261.pth
 100%| | 230M/230M [00:10<00:00, 23.3MB/s]

0.5 Set up training tools

```
[12]: from importlib import reload
reload(yolo_loss) # update the import if the implementation has changed

criterion = yolo_loss.YoloLoss(S, B, lambda_coord, lambda_noobj)
optimizer = torch.optim.SGD(net.parameters(), lr=learning_rate, momentum=0.9,
↪ weight_decay=5e-4)
```

0.6 Train detector

```
[19]: best_test_loss = np.inf
learning_rate = learning_rate
checkpoints_dir = './checkpoints_152'

os.makedirs(checkpoints_dir, exist_ok=True)
for epoch in range(num_epochs):
    net.train()

    # Update learning rate late in training
    if epoch == 30 or epoch == 40:
        learning_rate /= 10.0

    for param_group in optimizer.param_groups:
        param_group['lr'] = learning_rate

    print('\n\nStarting epoch %d / %d' % (epoch + 1, num_epochs))
    print('Learning Rate for this epoch: {}'.format(learning_rate))

    total_loss = collections.defaultdict(int)
```

```

start_iteration_time = time.time()
for i, data in enumerate(train_loader):
    data = (item.to(device) for item in data)
    images, target_boxes, target_cls, has_object_map = data
    pred = net(images)
    loss_dict = criterion(pred, target_boxes, target_cls, has_object_map)
    for key in loss_dict:
        total_loss[key] += loss_dict[key].item()

    optimizer.zero_grad()
    loss_dict['total_loss'].backward()
    optimizer.step()

    if (i+1) % 50 == 0:
        outstring = 'Epoch [%d/%d], Iter [%d/%d], Loss: ' % ((epoch+1,
↪num_epochs, i+1, len(train_loader)))
        outstring += ', '.join( "%s=%.3f" % (key[:-5], val / (i+1)) for
↪key, val in total_loss.items() )
        print(outstring)
    print('Time for epoch: %.2f seconds' % (time.time() - start_iteration_time))

# evaluate the network on the test data
if (epoch + 1) % 5 == 0:
    torch.save(net.state_dict(), f'{checkpoints_dir}/detector_epoch_%d.pth'
↪% (epoch+1))
    test_aps = evaluate(net, test_dataset_file=annotation_file_test,
↪img_root=file_root_test)
    print(epoch, test_aps)
    with torch.no_grad():
        test_loss = 0.0
        net.eval()
        for i, data in enumerate(test_loader):
            data = (item.to(device) for item in data)
            images, target_boxes, target_cls, has_object_map = data

            pred = net(images)
            loss_dict = criterion(pred, target_boxes, target_cls,
↪has_object_map)
            test_loss += loss_dict['total_loss'].item()
        test_loss /= len(test_loader)

    if best_test_loss > test_loss:
        best_test_loss = test_loss
        print('Updating best test loss: %.5f' % best_test_loss)
        torch.save(net.state_dict(), f'{checkpoints_dir}/best_detector.pth')

```

```
torch.save(net.state_dict(),f'{checkpoints_dir}/detector.pth')
```

Starting epoch 1 / 50

Learning Rate for this epoch: 0.001

Epoch [1/50], Iter [50/418], Loss: total=23.474, reg=45.999,
containing_obj=2.932, no_obj=150.138, cls=82.618
Epoch [1/50], Iter [100/418], Loss: total=16.074, reg=42.598,
containing_obj=4.184, no_obj=77.709, cls=68.396
Epoch [1/50], Iter [150/418], Loss: total=13.236, reg=39.787,
containing_obj=4.895, no_obj=53.031, cls=61.124
Epoch [1/50], Iter [200/418], Loss: total=11.454, reg=36.819,
containing_obj=5.362, no_obj=40.499, cls=54.771
Epoch [1/50], Iter [250/418], Loss: total=10.331, reg=35.158,
containing_obj=5.759, no_obj=32.884, cls=50.173
Epoch [1/50], Iter [300/418], Loss: total=9.609, reg=34.108,
containing_obj=6.152, no_obj=27.753, cls=47.290
Epoch [1/50], Iter [350/418], Loss: total=8.965, reg=32.805,
containing_obj=6.370, no_obj=24.055, cls=44.352
Epoch [1/50], Iter [400/418], Loss: total=8.499, reg=31.996,
containing_obj=6.545, no_obj=21.258, cls=42.192
Time for epoch: 42.69 seconds
Updating best test loss: 5.15525

Starting epoch 2 / 50

Learning Rate for this epoch: 0.001

Epoch [2/50], Iter [50/418], Loss: total=5.168, reg=26.537,
containing_obj=8.085, no_obj=1.502, cls=25.892
Epoch [2/50], Iter [100/418], Loss: total=4.938, reg=25.413,
containing_obj=8.206, no_obj=1.453, cls=24.180
Epoch [2/50], Iter [150/418], Loss: total=4.964, reg=25.845,
containing_obj=8.451, no_obj=1.410, cls=23.862
Epoch [2/50], Iter [200/418], Loss: total=4.902, reg=25.599,
containing_obj=8.504, no_obj=1.373, cls=23.350
Epoch [2/50], Iter [250/418], Loss: total=4.817, reg=25.017,
containing_obj=8.492, no_obj=1.342, cls=22.957
Epoch [2/50], Iter [300/418], Loss: total=4.786, reg=25.001,
containing_obj=8.530, no_obj=1.314, cls=22.586
Epoch [2/50], Iter [350/418], Loss: total=4.752, reg=24.908,
containing_obj=8.579, no_obj=1.292, cls=22.240
Epoch [2/50], Iter [400/418], Loss: total=4.696, reg=24.635,
containing_obj=8.563, no_obj=1.274, cls=21.883
Time for epoch: 42.00 seconds
Updating best test loss: 4.47245

Starting epoch 3 / 50

Learning Rate for this epoch: 0.001

Epoch [3/50], Iter [50/418], Loss: total=4.284, reg=22.606,
containing_obj=8.921, no_obj=1.151, cls=18.736
Epoch [3/50], Iter [100/418], Loss: total=4.324, reg=23.119,
containing_obj=8.984, no_obj=1.164, cls=18.615
Epoch [3/50], Iter [150/418], Loss: total=4.352, reg=23.310,
containing_obj=8.990, no_obj=1.179, cls=18.745
Epoch [3/50], Iter [200/418], Loss: total=4.329, reg=23.286,
containing_obj=8.910, no_obj=1.199, cls=18.549
Epoch [3/50], Iter [250/418], Loss: total=4.344, reg=23.435,
containing_obj=8.844, no_obj=1.220, cls=18.634
Epoch [3/50], Iter [300/418], Loss: total=4.314, reg=23.307,
containing_obj=8.830, no_obj=1.238, cls=18.398
Epoch [3/50], Iter [350/418], Loss: total=4.291, reg=23.138,
containing_obj=8.801, no_obj=1.252, cls=18.305
Epoch [3/50], Iter [400/418], Loss: total=4.283, reg=23.132,
containing_obj=8.795, no_obj=1.269, cls=18.205
Time for epoch: 41.41 seconds
Updating best test loss: 4.15479

Starting epoch 4 / 50

Learning Rate for this epoch: 0.001

Epoch [4/50], Iter [50/418], Loss: total=4.174, reg=22.450,
containing_obj=8.629, no_obj=1.463, cls=17.549
Epoch [4/50], Iter [100/418], Loss: total=4.133, reg=22.225,
containing_obj=8.678, no_obj=1.444, cls=17.246
Epoch [4/50], Iter [150/418], Loss: total=4.073, reg=21.994,
containing_obj=8.621, no_obj=1.428, cls=16.833
Epoch [4/50], Iter [200/418], Loss: total=4.042, reg=22.096,
containing_obj=8.619, no_obj=1.429, cls=16.362
Epoch [4/50], Iter [250/418], Loss: total=4.011, reg=21.947,
containing_obj=8.647, no_obj=1.430, cls=16.114
Epoch [4/50], Iter [300/418], Loss: total=3.964, reg=21.722,
containing_obj=8.550, no_obj=1.431, cls=15.871
Epoch [4/50], Iter [350/418], Loss: total=3.948, reg=21.678,
containing_obj=8.571, no_obj=1.424, cls=15.708
Epoch [4/50], Iter [400/418], Loss: total=3.915, reg=21.494,
containing_obj=8.507, no_obj=1.426, cls=15.553
Time for epoch: 42.10 seconds
Updating best test loss: 3.79224

Starting epoch 5 / 50

Learning Rate for this epoch: 0.001

Epoch [5/50], Iter [50/418], Loss: total=3.745, reg=21.085,

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containing_obj=8.536, no_obj=1.458, cls=13.858
Epoch [5/50], Iter [100/418], Loss: total=3.679, reg=20.390,
containing_obj=8.426, no_obj=1.473, cls=13.854
Epoch [5/50], Iter [150/418], Loss: total=3.673, reg=20.544,
containing_obj=8.365, no_obj=1.471, cls=13.691
Epoch [5/50], Iter [200/418], Loss: total=3.735, reg=21.049,
containing_obj=8.449, no_obj=1.455, cls=13.868
Epoch [5/50], Iter [250/418], Loss: total=3.705, reg=21.084,
containing_obj=8.455, no_obj=1.451, cls=13.465
Epoch [5/50], Iter [300/418], Loss: total=3.672, reg=20.905,
containing_obj=8.436, no_obj=1.443, cls=13.280
Epoch [5/50], Iter [350/418], Loss: total=3.649, reg=20.812,
containing_obj=8.449, no_obj=1.427, cls=13.105
Epoch [5/50], Iter [400/418], Loss: total=3.634, reg=20.798,
containing_obj=8.427, no_obj=1.429, cls=12.959
Time for epoch: 40.85 seconds
---Evaluate model on test samples---

100%|      | 4950/4950 [01:46<00:00, 46.34it/s]

---class aeroplane ap 0.26014848912511584---
---class bicycle ap 0.12288521921985074---
---class bird ap 0.21968523897765974---
---class boat ap 0.0--- (no predictions for this class)
---class bottle ap 0.035391445373407514---
---class bus ap 0.12469120405911886---
---class car ap 0.45763868746373415---
---class cat ap 0.2800353742908921---
---class chair ap 0.10852482316092332---
---class cow ap 0.004098360655737705---
---class diningtable ap 0.0--- (no predictions for this class)
---class dog ap 0.2158215836133348---
---class horse ap 0.3442628401160167---
---class motorbike ap 0.030059171597633133---
---class person ap 0.3027209390912654---
---class pottedplant ap 0.01962535004347119---
---class sheep ap 0.09406345110665365---
---class sofa ap 0.0--- (no predictions for this class)
---class train ap 0.27641589958195123---
---class tvmonitor ap 0.16786601331574713---
---map 0.15319670453962567---
4 [np.float64(0.26014848912511584), np.float64(0.12288521921985074),
np.float64(0.21968523897765974), 0.0, np.float64(0.035391445373407514),
np.float64(0.12469120405911886), np.float64(0.45763868746373415),
np.float64(0.2800353742908921), np.float64(0.10852482316092332),
np.float64(0.004098360655737705), 0.0, np.float64(0.2158215836133348),
np.float64(0.3442628401160167), np.float64(0.030059171597633133),
np.float64(0.3027209390912654), np.float64(0.01962535004347119),
np.float64(0.09406345110665365), 0.0, np.float64(0.27641589958195123),

```

```
np.float64(0.16786601331574713)]
Updating best test loss: 3.56206
```

```
Starting epoch 6 / 50
Learning Rate for this epoch: 0.001
Epoch [6/50], Iter [50/418], Loss: total=3.522, reg=19.998,
containing_obj=8.722, no_obj=1.479, cls=12.060
Epoch [6/50], Iter [100/418], Loss: total=3.437, reg=19.602,
containing_obj=8.441, no_obj=1.455, cls=11.741
Epoch [6/50], Iter [150/418], Loss: total=3.401, reg=19.466,
containing_obj=8.446, no_obj=1.447, cls=11.448
Epoch [6/50], Iter [200/418], Loss: total=3.393, reg=19.582,
containing_obj=8.484, no_obj=1.426, cls=11.229
Epoch [6/50], Iter [250/418], Loss: total=3.439, reg=19.965,
containing_obj=8.556, no_obj=1.431, cls=11.319
Epoch [6/50], Iter [300/418], Loss: total=3.408, reg=19.885,
containing_obj=8.475, no_obj=1.439, cls=11.093
Epoch [6/50], Iter [350/418], Loss: total=3.437, reg=20.045,
containing_obj=8.521, no_obj=1.442, cls=11.236
Epoch [6/50], Iter [400/418], Loss: total=3.438, reg=20.125,
containing_obj=8.497, no_obj=1.450, cls=11.179
Time for epoch: 41.75 seconds
Updating best test loss: 3.41255
```

```
Starting epoch 7 / 50
Learning Rate for this epoch: 0.001
Epoch [7/50], Iter [50/418], Loss: total=3.282, reg=19.469,
containing_obj=8.643, no_obj=1.449, cls=9.818
Epoch [7/50], Iter [100/418], Loss: total=3.242, reg=18.847,
containing_obj=8.572, no_obj=1.444, cls=10.040
Epoch [7/50], Iter [150/418], Loss: total=3.290, reg=19.125,
containing_obj=8.673, no_obj=1.478, cls=10.201
Epoch [7/50], Iter [200/418], Loss: total=3.225, reg=18.852,
containing_obj=8.466, no_obj=1.481, cls=9.904
Epoch [7/50], Iter [250/418], Loss: total=3.234, reg=18.886,
containing_obj=8.482, no_obj=1.485, cls=9.960
Epoch [7/50], Iter [300/418], Loss: total=3.264, reg=19.111,
containing_obj=8.539, no_obj=1.478, cls=10.035
Epoch [7/50], Iter [350/418], Loss: total=3.269, reg=19.217,
containing_obj=8.568, no_obj=1.485, cls=9.960
Epoch [7/50], Iter [400/418], Loss: total=3.260, reg=19.133,
containing_obj=8.604, no_obj=1.476, cls=9.905
Time for epoch: 41.88 seconds
Updating best test loss: 3.24609
```

Starting epoch 8 / 50

Learning Rate for this epoch: 0.001

Epoch [8/50], Iter [50/418], Loss: total=3.251, reg=19.342,
containing_obj=8.846, no_obj=1.497, cls=9.332

Epoch [8/50], Iter [100/418], Loss: total=3.105, reg=18.525,
containing_obj=8.637, no_obj=1.449, cls=8.647

Epoch [8/50], Iter [150/418], Loss: total=3.129, reg=18.643,
containing_obj=8.756, no_obj=1.447, cls=8.708

Epoch [8/50], Iter [200/418], Loss: total=3.151, reg=18.721,
containing_obj=8.770, no_obj=1.453, cls=8.868

Epoch [8/50], Iter [250/418], Loss: total=3.144, reg=18.708,
containing_obj=8.857, no_obj=1.437, cls=8.726

Epoch [8/50], Iter [300/418], Loss: total=3.130, reg=18.612,
containing_obj=8.880, no_obj=1.447, cls=8.620

Epoch [8/50], Iter [350/418], Loss: total=3.106, reg=18.461,
containing_obj=8.811, no_obj=1.453, cls=8.551

Epoch [8/50], Iter [400/418], Loss: total=3.110, reg=18.508,
containing_obj=8.801, no_obj=1.453, cls=8.552

Time for epoch: 41.64 seconds

Updating best test loss: 3.11589

Starting epoch 9 / 50

Learning Rate for this epoch: 0.001

Epoch [9/50], Iter [50/418], Loss: total=2.901, reg=17.912,
containing_obj=8.349, no_obj=1.452, cls=7.105

Epoch [9/50], Iter [100/418], Loss: total=2.996, reg=18.042,
containing_obj=8.857, no_obj=1.456, cls=7.601

Epoch [9/50], Iter [150/418], Loss: total=2.994, reg=17.979,
containing_obj=8.838, no_obj=1.456, cls=7.653

Epoch [9/50], Iter [200/418], Loss: total=2.970, reg=17.743,
containing_obj=8.817, no_obj=1.462, cls=7.616

Epoch [9/50], Iter [250/418], Loss: total=2.979, reg=17.783,
containing_obj=8.772, no_obj=1.479, cls=7.715

Epoch [9/50], Iter [300/418], Loss: total=2.975, reg=17.731,
containing_obj=8.846, no_obj=1.478, cls=7.646

Epoch [9/50], Iter [350/418], Loss: total=2.967, reg=17.689,
containing_obj=8.856, no_obj=1.476, cls=7.588

Epoch [9/50], Iter [400/418], Loss: total=2.953, reg=17.664,
containing_obj=8.830, no_obj=1.483, cls=7.455

Time for epoch: 42.48 seconds

Starting epoch 10 / 50

Learning Rate for this epoch: 0.001

Epoch [10/50], Iter [50/418], Loss: total=2.890, reg=17.285,
containing_obj=8.807, no_obj=1.523, cls=7.066

Epoch [10/50], Iter [100/418], Loss: total=2.900, reg=17.465,

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containing_obj=8.840, no_obj=1.525, cls=6.976
Epoch [10/50], Iter [150/418], Loss: total=2.915, reg=17.509,
containing_obj=8.954, no_obj=1.540, cls=6.982
Epoch [10/50], Iter [200/418], Loss: total=2.905, reg=17.460,
containing_obj=8.924, no_obj=1.549, cls=6.930
Epoch [10/50], Iter [250/418], Loss: total=2.872, reg=17.104,
containing_obj=8.875, no_obj=1.541, cls=6.949
Epoch [10/50], Iter [300/418], Loss: total=2.854, reg=16.989,
containing_obj=8.784, no_obj=1.537, cls=6.939
Epoch [10/50], Iter [350/418], Loss: total=2.860, reg=17.101,
containing_obj=8.800, no_obj=1.531, cls=6.890
Epoch [10/50], Iter [400/418], Loss: total=2.901, reg=17.353,
containing_obj=8.855, no_obj=1.537, cls=7.067
Time for epoch: 42.21 seconds
---Evaluate model on test samples---

100%|          | 4950/4950 [01:48<00:00, 45.52it/s]

---class aeroplane ap 0.4455839195063399---
---class bicycle ap 0.5007273847932744---
---class bird ap 0.34688284448777207---
---class boat ap 0.17024355682401338---
---class bottle ap 0.05140618076351904---
---class bus ap 0.4578087719553065---
---class car ap 0.5877565895556509---
---class cat ap 0.5520124511103224---
---class chair ap 0.2044201686122676---
---class cow ap 0.14156821537824205---
---class diningtable ap 0.0---
---class dog ap 0.42820195280100215---
---class horse ap 0.5246775440284173---
---class motorbike ap 0.44747515494789564---
---class person ap 0.41715692371982455---
---class pottedplant ap 0.09037552754836105---
---class sheep ap 0.28030096132819043---
---class sofa ap 0.2219713568590102---
---class train ap 0.49464518369165233---
---class tvmonitor ap 0.34859228599014874---
---map 0.33559034869506055---
9 [np.float64(0.4455839195063399), np.float64(0.5007273847932744),
np.float64(0.34688284448777207), np.float64(0.17024355682401338),
np.float64(0.05140618076351904), np.float64(0.4578087719553065),
np.float64(0.5877565895556509), np.float64(0.5520124511103224),
np.float64(0.2044201686122676), np.float64(0.14156821537824205),
np.float64(0.0), np.float64(0.42820195280100215),
np.float64(0.5246775440284173), np.float64(0.44747515494789564),
np.float64(0.41715692371982455), np.float64(0.09037552754836105),
np.float64(0.28030096132819043), np.float64(0.2219713568590102),
np.float64(0.49464518369165233), np.float64(0.34859228599014874)]

```

Updating best test loss: 3.02590

Starting epoch 11 / 50

Learning Rate for this epoch: 0.001

Epoch [11/50], Iter [50/418], Loss: total=2.746, reg=15.792,
containing_obj=8.889, no_obj=1.686, cls=6.584
Epoch [11/50], Iter [100/418], Loss: total=2.630, reg=15.493,
containing_obj=8.515, no_obj=1.654, cls=5.904
Epoch [11/50], Iter [150/418], Loss: total=2.657, reg=15.659,
containing_obj=8.622, no_obj=1.632, cls=5.969
Epoch [11/50], Iter [200/418], Loss: total=2.717, reg=16.042,
containing_obj=8.785, no_obj=1.603, cls=6.175
Epoch [11/50], Iter [250/418], Loss: total=2.723, reg=16.117,
containing_obj=8.748, no_obj=1.605, cls=6.201
Epoch [11/50], Iter [300/418], Loss: total=2.750, reg=16.432,
containing_obj=8.764, no_obj=1.614, cls=6.187
Epoch [11/50], Iter [350/418], Loss: total=2.749, reg=16.480,
containing_obj=8.765, no_obj=1.617, cls=6.124
Epoch [11/50], Iter [400/418], Loss: total=2.748, reg=16.456,
containing_obj=8.724, no_obj=1.646, cls=6.156
Time for epoch: 42.66 seconds
Updating best test loss: 2.96127

Starting epoch 12 / 50

Learning Rate for this epoch: 0.001

Epoch [12/50], Iter [50/418], Loss: total=2.667, reg=15.992,
containing_obj=8.431, no_obj=1.722, cls=5.863
Epoch [12/50], Iter [100/418], Loss: total=2.654, reg=16.085,
containing_obj=8.161, no_obj=1.798, cls=5.800
Epoch [12/50], Iter [150/418], Loss: total=2.645, reg=16.107,
containing_obj=8.250, no_obj=1.778, cls=5.607
Epoch [12/50], Iter [200/418], Loss: total=2.676, reg=16.306,
containing_obj=8.325, no_obj=1.786, cls=5.696
Epoch [12/50], Iter [250/418], Loss: total=2.661, reg=16.183,
containing_obj=8.329, no_obj=1.789, cls=5.631
Epoch [12/50], Iter [300/418], Loss: total=2.677, reg=16.332,
containing_obj=8.380, no_obj=1.785, cls=5.627
Epoch [12/50], Iter [350/418], Loss: total=2.675, reg=16.281,
containing_obj=8.368, no_obj=1.810, cls=5.643
Epoch [12/50], Iter [400/418], Loss: total=2.686, reg=16.332,
containing_obj=8.401, no_obj=1.826, cls=5.669
Time for epoch: 41.42 seconds
Updating best test loss: 2.92753

Starting epoch 13 / 50

Learning Rate for this epoch: 0.001
Epoch [13/50], Iter [50/418], Loss: total=2.674, reg=16.321,
containing_obj=8.508, no_obj=1.857, cls=5.397
Epoch [13/50], Iter [100/418], Loss: total=2.574, reg=15.728,
containing_obj=8.069, no_obj=1.900, cls=5.193
Epoch [13/50], Iter [150/418], Loss: total=2.590, reg=15.707,
containing_obj=8.018, no_obj=1.910, cls=5.444
Epoch [13/50], Iter [200/418], Loss: total=2.617, reg=15.925,
containing_obj=8.165, no_obj=1.907, cls=5.402
Epoch [13/50], Iter [250/418], Loss: total=2.637, reg=16.039,
containing_obj=8.221, no_obj=1.897, cls=5.482
Epoch [13/50], Iter [300/418], Loss: total=2.627, reg=15.969,
containing_obj=8.165, no_obj=1.921, cls=5.465
Epoch [13/50], Iter [350/418], Loss: total=2.622, reg=15.999,
containing_obj=8.149, no_obj=1.928, cls=5.383
Epoch [13/50], Iter [400/418], Loss: total=2.614, reg=15.945,
containing_obj=8.132, no_obj=1.945, cls=5.349
Time for epoch: 40.99 seconds
Updating best test loss: 2.83592

Starting epoch 14 / 50
Learning Rate for this epoch: 0.001
Epoch [14/50], Iter [50/418], Loss: total=2.530, reg=15.300,
containing_obj=8.126, no_obj=1.922, cls=5.009
Epoch [14/50], Iter [100/418], Loss: total=2.515, reg=15.090,
containing_obj=8.180, no_obj=1.951, cls=4.958
Epoch [14/50], Iter [150/418], Loss: total=2.506, reg=15.143,
containing_obj=8.110, no_obj=1.969, cls=4.849
Epoch [14/50], Iter [200/418], Loss: total=2.544, reg=15.448,
containing_obj=8.221, no_obj=1.982, cls=4.874
Epoch [14/50], Iter [250/418], Loss: total=2.560, reg=15.546,
containing_obj=8.185, no_obj=1.991, cls=5.004
Epoch [14/50], Iter [300/418], Loss: total=2.554, reg=15.454,
containing_obj=8.124, no_obj=2.009, cls=5.058
Epoch [14/50], Iter [350/418], Loss: total=2.548, reg=15.445,
containing_obj=8.084, no_obj=2.004, cls=5.038
Epoch [14/50], Iter [400/418], Loss: total=2.549, reg=15.476,
containing_obj=8.124, no_obj=1.993, cls=4.999
Time for epoch: 41.70 seconds
Updating best test loss: 2.83485

Starting epoch 15 / 50
Learning Rate for this epoch: 0.001
Epoch [15/50], Iter [50/418], Loss: total=2.612, reg=16.306,
containing_obj=8.350, no_obj=2.018, cls=4.668
Epoch [15/50], Iter [100/418], Loss: total=2.485, reg=15.300,

```

containing_obj=7.935, no_obj=2.006, cls=4.580
Epoch [15/50], Iter [150/418], Loss: total=2.483, reg=15.127,
containing_obj=7.978, no_obj=2.022, cls=4.668
Epoch [15/50], Iter [200/418], Loss: total=2.471, reg=15.075,
containing_obj=7.996, no_obj=2.008, cls=4.567
Epoch [15/50], Iter [250/418], Loss: total=2.470, reg=15.064,
containing_obj=7.998, no_obj=2.013, cls=4.565
Epoch [15/50], Iter [300/418], Loss: total=2.473, reg=15.066,
containing_obj=8.017, no_obj=2.018, cls=4.575
Epoch [15/50], Iter [350/418], Loss: total=2.448, reg=14.893,
containing_obj=7.910, no_obj=2.033, cls=4.543
Epoch [15/50], Iter [400/418], Loss: total=2.468, reg=15.019,
containing_obj=8.004, no_obj=2.014, cls=4.575
Time for epoch: 41.09 seconds
---Evaluate model on test samples---

100%|          | 4950/4950 [01:52<00:00, 44.00it/s]

---class aeroplane ap 0.38961253214983305---
---class bicycle ap 0.5808368067026122---
---class bird ap 0.3669519282202006---
---class boat ap 0.20456525086064645---
---class bottle ap 0.13072977785476986---
---class bus ap 0.5111497794330722---
---class car ap 0.5847470191942881---
---class cat ap 0.6240404050067814---
---class chair ap 0.24917637669486065---
---class cow ap 0.27355400862291024---
---class diningtable ap 0.045160341276846136---
---class dog ap 0.5419328424256288---
---class horse ap 0.5755482964879948---
---class motorbike ap 0.4345929524501425---
---class person ap 0.4561705094114411---
---class pottedplant ap 0.10065213499732598---
---class sheep ap 0.3905338756810488---
---class sofa ap 0.36336812861318474---
---class train ap 0.5227373198009446---
---class tvmonitor ap 0.43602295799381646---
---map 0.3891041621939174---
14 [np.float64(0.38961253214983305), np.float64(0.5808368067026122),
np.float64(0.3669519282202006), np.float64(0.20456525086064645),
np.float64(0.13072977785476986), np.float64(0.5111497794330722),
np.float64(0.5847470191942881), np.float64(0.6240404050067814),
np.float64(0.24917637669486065), np.float64(0.27355400862291024),
np.float64(0.045160341276846136), np.float64(0.5419328424256288),
np.float64(0.5755482964879948), np.float64(0.4345929524501425),
np.float64(0.4561705094114411), np.float64(0.10065213499732598),
np.float64(0.3905338756810488), np.float64(0.36336812861318474),
np.float64(0.5227373198009446), np.float64(0.43602295799381646)]

```

Updating best test loss: 2.78544

Starting epoch 16 / 50

Learning Rate for this epoch: 0.001

Epoch [16/50], Iter [50/418], Loss: total=2.550, reg=15.405,
containing_obj=8.542, no_obj=2.121, cls=4.537
Epoch [16/50], Iter [100/418], Loss: total=2.434, reg=14.864,
containing_obj=7.940, no_obj=2.110, cls=4.298
Epoch [16/50], Iter [150/418], Loss: total=2.440, reg=14.845,
containing_obj=7.926, no_obj=2.091, cls=4.421
Epoch [16/50], Iter [200/418], Loss: total=2.390, reg=14.550,
containing_obj=7.781, no_obj=2.095, cls=4.249
Epoch [16/50], Iter [250/418], Loss: total=2.392, reg=14.502,
containing_obj=7.839, no_obj=2.084, cls=4.283
Epoch [16/50], Iter [300/418], Loss: total=2.404, reg=14.578,
containing_obj=7.906, no_obj=2.076, cls=4.291
Epoch [16/50], Iter [350/418], Loss: total=2.414, reg=14.650,
containing_obj=7.929, no_obj=2.080, cls=4.303
Epoch [16/50], Iter [400/418], Loss: total=2.392, reg=14.476,
containing_obj=7.863, no_obj=2.078, cls=4.285
Time for epoch: 41.21 seconds
Updating best test loss: 2.75168

Starting epoch 17 / 50

Learning Rate for this epoch: 0.001

Epoch [17/50], Iter [50/418], Loss: total=2.404, reg=14.308,
containing_obj=7.907, no_obj=2.039, cls=4.598
Epoch [17/50], Iter [100/418], Loss: total=2.285, reg=13.645,
containing_obj=7.516, no_obj=2.039, cls=4.223
Epoch [17/50], Iter [150/418], Loss: total=2.347, reg=14.019,
containing_obj=7.790, no_obj=2.013, cls=4.345
Epoch [17/50], Iter [200/418], Loss: total=2.325, reg=13.931,
containing_obj=7.760, no_obj=2.033, cls=4.171
Epoch [17/50], Iter [250/418], Loss: total=2.309, reg=13.789,
containing_obj=7.713, no_obj=2.047, cls=4.161
Epoch [17/50], Iter [300/418], Loss: total=2.332, reg=14.005,
containing_obj=7.829, no_obj=2.039, cls=4.108
Epoch [17/50], Iter [350/418], Loss: total=2.353, reg=14.162,
containing_obj=7.896, no_obj=2.043, cls=4.132
Epoch [17/50], Iter [400/418], Loss: total=2.343, reg=14.105,
containing_obj=7.844, no_obj=2.051, cls=4.114
Time for epoch: 41.48 seconds

Starting epoch 18 / 50

Learning Rate for this epoch: 0.001

Epoch [18/50], Iter [50/418], Loss: total=2.370, reg=13.848,
containing_obj=8.049, no_obj=2.029, cls=4.514
Epoch [18/50], Iter [100/418], Loss: total=2.313, reg=13.807,
containing_obj=7.940, no_obj=2.033, cls=3.977
Epoch [18/50], Iter [150/418], Loss: total=2.318, reg=13.893,
containing_obj=7.862, no_obj=2.049, cls=4.016
Epoch [18/50], Iter [200/418], Loss: total=2.315, reg=13.866,
containing_obj=7.877, no_obj=2.057, cls=3.976
Epoch [18/50], Iter [250/418], Loss: total=2.299, reg=13.752,
containing_obj=7.837, no_obj=2.050, cls=3.953
Epoch [18/50], Iter [300/418], Loss: total=2.283, reg=13.691,
containing_obj=7.742, no_obj=2.043, cls=3.925
Epoch [18/50], Iter [350/418], Loss: total=2.290, reg=13.739,
containing_obj=7.762, no_obj=2.057, cls=3.921
Epoch [18/50], Iter [400/418], Loss: total=2.304, reg=13.884,
containing_obj=7.747, no_obj=2.062, cls=3.955
Time for epoch: 42.36 seconds
Updating best test loss: 2.69762

Starting epoch 19 / 50

Learning Rate for this epoch: 0.001

Epoch [19/50], Iter [50/418], Loss: total=2.259, reg=13.656,
containing_obj=7.809, no_obj=2.258, cls=3.388
Epoch [19/50], Iter [100/418], Loss: total=2.276, reg=13.539,
containing_obj=7.734, no_obj=2.165, cls=3.876
Epoch [19/50], Iter [150/418], Loss: total=2.285, reg=13.609,
containing_obj=7.769, no_obj=2.169, cls=3.867
Epoch [19/50], Iter [200/418], Loss: total=2.256, reg=13.530,
containing_obj=7.666, no_obj=2.154, cls=3.727
Epoch [19/50], Iter [250/418], Loss: total=2.246, reg=13.497,
containing_obj=7.611, no_obj=2.139, cls=3.706
Epoch [19/50], Iter [300/418], Loss: total=2.241, reg=13.458,
containing_obj=7.591, no_obj=2.136, cls=3.706
Epoch [19/50], Iter [350/418], Loss: total=2.257, reg=13.539,
containing_obj=7.636, no_obj=2.149, cls=3.762
Epoch [19/50], Iter [400/418], Loss: total=2.270, reg=13.625,
containing_obj=7.662, no_obj=2.145, cls=3.815
Time for epoch: 41.58 seconds
Updating best test loss: 2.69565

Starting epoch 20 / 50

Learning Rate for this epoch: 0.001

Epoch [20/50], Iter [50/418], Loss: total=2.147, reg=13.364,
containing_obj=7.125, no_obj=2.148, cls=3.124
Epoch [20/50], Iter [100/418], Loss: total=2.176, reg=13.310,
containing_obj=7.402, no_obj=2.181, cls=3.222

```

Epoch [20/50], Iter [150/418], Loss: total=2.196, reg=13.448,
containing_obj=7.504, no_obj=2.167, cls=3.231
Epoch [20/50], Iter [200/418], Loss: total=2.196, reg=13.317,
containing_obj=7.458, no_obj=2.173, cls=3.403
Epoch [20/50], Iter [250/418], Loss: total=2.204, reg=13.346,
containing_obj=7.465, no_obj=2.160, cls=3.474
Epoch [20/50], Iter [300/418], Loss: total=2.193, reg=13.242,
containing_obj=7.502, no_obj=2.165, cls=3.404
Epoch [20/50], Iter [350/418], Loss: total=2.212, reg=13.401,
containing_obj=7.505, no_obj=2.176, cls=3.461
Epoch [20/50], Iter [400/418], Loss: total=2.217, reg=13.434,
containing_obj=7.500, no_obj=2.164, cls=3.500
Time for epoch: 41.09 seconds
---Evaluate model on test samples---

100%|          | 4950/4950 [01:51<00:00, 44.22it/s]

---class aeroplane ap 0.5073530470406515---
---class bicycle ap 0.6201752205640809---
---class bird ap 0.4551376007245085---
---class boat ap 0.28988266988104683---
---class bottle ap 0.1674335036623753---
---class bus ap 0.6038733817740145---
---class car ap 0.6556696784742349---
---class cat ap 0.7096721995010875---
---class chair ap 0.28101747093196316---
---class cow ap 0.44265347071450123---
---class diningtable ap 0.2643820419790872---
---class dog ap 0.6137590724533485---
---class horse ap 0.6534375359786622---
---class motorbike ap 0.5077807961090144---
---class person ap 0.48932966724118737---
---class pottedplant ap 0.20205392286411622---
---class sheep ap 0.44588398523851075---
---class sofa ap 0.42652045766155344---
---class train ap 0.6654287385682736---
---class tvmonitor ap 0.4568233952769335---
---map 0.4729133928319575---
19 [np.float64(0.5073530470406515), np.float64(0.6201752205640809),
np.float64(0.4551376007245085), np.float64(0.28988266988104683),
np.float64(0.1674335036623753), np.float64(0.6038733817740145),
np.float64(0.6556696784742349), np.float64(0.7096721995010875),
np.float64(0.28101747093196316), np.float64(0.44265347071450123),
np.float64(0.2643820419790872), np.float64(0.6137590724533485),
np.float64(0.6534375359786622), np.float64(0.5077807961090144),
np.float64(0.48932966724118737), np.float64(0.20205392286411622),
np.float64(0.44588398523851075), np.float64(0.42652045766155344),
np.float64(0.6654287385682736), np.float64(0.4568233952769335)]
Updating best test loss: 2.67134

```

Starting epoch 21 / 50

Learning Rate for this epoch: 0.001

Epoch [21/50], Iter [50/418], Loss: total=2.181, reg=13.046,
containing_obj=7.561, no_obj=2.204, cls=3.356
Epoch [21/50], Iter [100/418], Loss: total=2.171, reg=12.879,
containing_obj=7.537, no_obj=2.184, cls=3.448
Epoch [21/50], Iter [150/418], Loss: total=2.195, reg=13.241,
containing_obj=7.455, no_obj=2.173, cls=3.466
Epoch [21/50], Iter [200/418], Loss: total=2.170, reg=13.093,
containing_obj=7.407, no_obj=2.189, cls=3.355
Epoch [21/50], Iter [250/418], Loss: total=2.163, reg=13.032,
containing_obj=7.452, no_obj=2.160, cls=3.317
Epoch [21/50], Iter [300/418], Loss: total=2.175, reg=13.055,
containing_obj=7.502, no_obj=2.179, cls=3.361
Epoch [21/50], Iter [350/418], Loss: total=2.162, reg=13.005,
containing_obj=7.445, no_obj=2.180, cls=3.309
Epoch [21/50], Iter [400/418], Loss: total=2.156, reg=12.961,
containing_obj=7.450, no_obj=2.171, cls=3.290
Time for epoch: 41.89 seconds

Starting epoch 22 / 50

Learning Rate for this epoch: 0.001

Epoch [22/50], Iter [50/418], Loss: total=2.192, reg=13.658,
containing_obj=7.264, no_obj=2.207, cls=3.175
Epoch [22/50], Iter [100/418], Loss: total=2.200, reg=13.429,
containing_obj=7.347, no_obj=2.264, cls=3.357
Epoch [22/50], Iter [150/418], Loss: total=2.216, reg=13.522,
containing_obj=7.433, no_obj=2.196, cls=3.439
Epoch [22/50], Iter [200/418], Loss: total=2.173, reg=13.300,
containing_obj=7.256, no_obj=2.232, cls=3.289
Epoch [22/50], Iter [250/418], Loss: total=2.163, reg=13.174,
containing_obj=7.338, no_obj=2.201, cls=3.240
Epoch [22/50], Iter [300/418], Loss: total=2.118, reg=12.850,
containing_obj=7.180, no_obj=2.214, cls=3.171
Epoch [22/50], Iter [350/418], Loss: total=2.145, reg=13.044,
containing_obj=7.312, no_obj=2.212, cls=3.172
Epoch [22/50], Iter [400/418], Loss: total=2.154, reg=13.143,
containing_obj=7.313, no_obj=2.206, cls=3.191
Time for epoch: 41.98 seconds
Updating best test loss: 2.62460

Starting epoch 23 / 50

Learning Rate for this epoch: 0.001

Epoch [23/50], Iter [50/418], Loss: total=2.156, reg=12.911,

containing_obj=7.605, no_obj=2.191, cls=3.160
Epoch [23/50], Iter [100/418], Loss: total=2.101, reg=12.720,
containing_obj=7.376, no_obj=2.158, cls=2.954
Epoch [23/50], Iter [150/418], Loss: total=2.094, reg=12.518,
containing_obj=7.519, no_obj=2.141, cls=2.947
Epoch [23/50], Iter [200/418], Loss: total=2.086, reg=12.526,
containing_obj=7.405, no_obj=2.182, cls=2.916
Epoch [23/50], Iter [250/418], Loss: total=2.066, reg=12.354,
containing_obj=7.332, no_obj=2.200, cls=2.907
Epoch [23/50], Iter [300/418], Loss: total=2.073, reg=12.343,
containing_obj=7.385, no_obj=2.215, cls=2.931
Epoch [23/50], Iter [350/418], Loss: total=2.071, reg=12.356,
containing_obj=7.376, no_obj=2.198, cls=2.919
Epoch [23/50], Iter [400/418], Loss: total=2.075, reg=12.366,
containing_obj=7.383, no_obj=2.193, cls=2.955
Time for epoch: 42.13 seconds
Updating best test loss: 2.61639

Starting epoch 24 / 50
Learning Rate for this epoch: 0.001
Epoch [24/50], Iter [50/418], Loss: total=2.003, reg=11.832,
containing_obj=7.037, no_obj=2.258, cls=2.906
Epoch [24/50], Iter [100/418], Loss: total=2.051, reg=12.499,
containing_obj=7.141, no_obj=2.178, cls=2.797
Epoch [24/50], Iter [150/418], Loss: total=2.078, reg=12.443,
containing_obj=7.427, no_obj=2.217, cls=2.851
Epoch [24/50], Iter [200/418], Loss: total=2.102, reg=12.731,
containing_obj=7.399, no_obj=2.241, cls=2.856
Epoch [24/50], Iter [250/418], Loss: total=2.098, reg=12.643,
containing_obj=7.387, no_obj=2.239, cls=2.912
Epoch [24/50], Iter [300/418], Loss: total=2.077, reg=12.445,
containing_obj=7.323, no_obj=2.227, cls=2.930
Epoch [24/50], Iter [350/418], Loss: total=2.081, reg=12.459,
containing_obj=7.317, no_obj=2.246, cls=2.945
Epoch [24/50], Iter [400/418], Loss: total=2.084, reg=12.472,
containing_obj=7.320, no_obj=2.229, cls=2.982
Time for epoch: 42.21 seconds
Updating best test loss: 2.61095

Starting epoch 25 / 50
Learning Rate for this epoch: 0.001
Epoch [25/50], Iter [50/418], Loss: total=1.943, reg=11.996,
containing_obj=6.666, no_obj=2.150, cls=2.504
Epoch [25/50], Iter [100/418], Loss: total=1.939, reg=11.891,
containing_obj=6.759, no_obj=2.181, cls=2.435
Epoch [25/50], Iter [150/418], Loss: total=1.978, reg=12.199,

```

containing_obj=6.871, no_obj=2.163, cls=2.506
Epoch [25/50], Iter [200/418], Loss: total=2.017, reg=12.434,
containing_obj=6.991, no_obj=2.181, cls=2.599
Epoch [25/50], Iter [250/418], Loss: total=2.071, reg=12.613,
containing_obj=7.282, no_obj=2.191, cls=2.771
Epoch [25/50], Iter [300/418], Loss: total=2.052, reg=12.462,
containing_obj=7.198, no_obj=2.209, cls=2.756
Epoch [25/50], Iter [350/418], Loss: total=2.056, reg=12.505,
containing_obj=7.185, no_obj=2.221, cls=2.767
Epoch [25/50], Iter [400/418], Loss: total=2.051, reg=12.494,
containing_obj=7.121, no_obj=2.220, cls=2.771
Time for epoch: 42.13 seconds
---Evaluate model on test samples---
100%|      | 4950/4950 [01:51<00:00, 44.44it/s]

---class aeroplane ap 0.524530620624914---
---class bicycle ap 0.6291677623582208---
---class bird ap 0.5103411214016357---
---class boat ap 0.3061612691393525---
---class bottle ap 0.19613403944250446---
---class bus ap 0.5783606336463101---
---class car ap 0.6568390586561228---
---class cat ap 0.7027748644486028---
---class chair ap 0.3063308516924699---
---class cow ap 0.45529854059168384---
---class diningtable ap 0.25312732608737964---
---class dog ap 0.6267945598906639---
---class horse ap 0.5993794609644638---
---class motorbike ap 0.48221720756485886---
---class person ap 0.5492897845567772---
---class pottedplant ap 0.20434343458881976---
---class sheep ap 0.5076815863569948---
---class sofa ap 0.47078491215122586---
---class train ap 0.5926340962129504---
---class tvmonitor ap 0.4898080816755614---
---map 0.48209996060257565---
24 [np.float64(0.524530620624914), np.float64(0.6291677623582208),
np.float64(0.5103411214016357), np.float64(0.3061612691393525),
np.float64(0.19613403944250446), np.float64(0.5783606336463101),
np.float64(0.6568390586561228), np.float64(0.7027748644486028),
np.float64(0.3063308516924699), np.float64(0.45529854059168384),
np.float64(0.25312732608737964), np.float64(0.6267945598906639),
np.float64(0.5993794609644638), np.float64(0.48221720756485886),
np.float64(0.5492897845567772), np.float64(0.20434343458881976),
np.float64(0.5076815863569948), np.float64(0.47078491215122586),
np.float64(0.5926340962129504), np.float64(0.4898080816755614)]
Updating best test loss: 2.59928

```

Starting epoch 26 / 50

Learning Rate for this epoch: 0.001

Epoch [26/50], Iter [50/418], Loss: total=1.909, reg=11.315,
containing_obj=6.952, no_obj=2.150, cls=2.496
Epoch [26/50], Iter [100/418], Loss: total=1.976, reg=11.688,
containing_obj=7.130, no_obj=2.293, cls=2.600
Epoch [26/50], Iter [150/418], Loss: total=1.982, reg=11.774,
containing_obj=7.181, no_obj=2.265, cls=2.567
Epoch [26/50], Iter [200/418], Loss: total=1.983, reg=11.846,
containing_obj=7.120, no_obj=2.280, cls=2.554
Epoch [26/50], Iter [250/418], Loss: total=1.975, reg=11.812,
containing_obj=7.071, no_obj=2.272, cls=2.549
Epoch [26/50], Iter [300/418], Loss: total=1.982, reg=11.833,
containing_obj=7.080, no_obj=2.281, cls=2.586
Epoch [26/50], Iter [350/418], Loss: total=1.975, reg=11.717,
containing_obj=7.063, no_obj=2.263, cls=2.654
Epoch [26/50], Iter [400/418], Loss: total=1.972, reg=11.679,
containing_obj=7.092, no_obj=2.266, cls=2.631
Time for epoch: 41.27 seconds

Starting epoch 27 / 50

Learning Rate for this epoch: 0.001

Epoch [27/50], Iter [50/418], Loss: total=1.988, reg=12.000,
containing_obj=7.415, no_obj=2.191, cls=2.247
Epoch [27/50], Iter [100/418], Loss: total=1.977, reg=11.838,
containing_obj=7.226, no_obj=2.218, cls=2.446
Epoch [27/50], Iter [150/418], Loss: total=1.970, reg=11.730,
containing_obj=7.144, no_obj=2.209, cls=2.558
Epoch [27/50], Iter [200/418], Loss: total=1.971, reg=11.642,
containing_obj=7.179, no_obj=2.221, cls=2.610
Epoch [27/50], Iter [250/418], Loss: total=1.990, reg=11.696,
containing_obj=7.232, no_obj=2.254, cls=2.695
Epoch [27/50], Iter [300/418], Loss: total=1.982, reg=11.700,
containing_obj=7.200, no_obj=2.238, cls=2.642
Epoch [27/50], Iter [350/418], Loss: total=1.969, reg=11.601,
containing_obj=7.196, no_obj=2.220, cls=2.612
Epoch [27/50], Iter [400/418], Loss: total=1.959, reg=11.580,
containing_obj=7.131, no_obj=2.217, cls=2.585
Time for epoch: 41.95 seconds
Updating best test loss: 2.58700

Starting epoch 28 / 50

Learning Rate for this epoch: 0.001

Epoch [28/50], Iter [50/418], Loss: total=2.072, reg=12.508,
containing_obj=7.419, no_obj=2.328, cls=2.605

Epoch [28/50], Iter [100/418], Loss: total=1.994, reg=11.882,
containing_obj=7.065, no_obj=2.352, cls=2.624
Epoch [28/50], Iter [150/418], Loss: total=1.966, reg=11.687,
containing_obj=7.191, no_obj=2.311, cls=2.402
Epoch [28/50], Iter [200/418], Loss: total=1.979, reg=11.812,
containing_obj=7.209, no_obj=2.296, cls=2.427
Epoch [28/50], Iter [250/418], Loss: total=1.977, reg=11.786,
containing_obj=7.196, no_obj=2.285, cls=2.453
Epoch [28/50], Iter [300/418], Loss: total=1.964, reg=11.731,
containing_obj=7.169, no_obj=2.267, cls=2.401
Epoch [28/50], Iter [350/418], Loss: total=1.941, reg=11.531,
containing_obj=7.105, no_obj=2.264, cls=2.394
Epoch [28/50], Iter [400/418], Loss: total=1.936, reg=11.485,
containing_obj=7.076, no_obj=2.271, cls=2.397
Time for epoch: 41.69 seconds
Updating best test loss: 2.58613

Starting epoch 29 / 50

Learning Rate for this epoch: 0.001

Epoch [29/50], Iter [50/418], Loss: total=1.822, reg=10.653,
containing_obj=6.720, no_obj=2.247, cls=2.244
Epoch [29/50], Iter [100/418], Loss: total=1.878, reg=11.010,
containing_obj=6.976, no_obj=2.216, cls=2.335
Epoch [29/50], Iter [150/418], Loss: total=1.874, reg=10.996,
containing_obj=6.825, no_obj=2.299, cls=2.368
Epoch [29/50], Iter [200/418], Loss: total=1.885, reg=11.058,
containing_obj=6.909, no_obj=2.268, cls=2.381
Epoch [29/50], Iter [250/418], Loss: total=1.877, reg=11.005,
containing_obj=6.935, no_obj=2.260, cls=2.327
Epoch [29/50], Iter [300/418], Loss: total=1.903, reg=11.172,
containing_obj=6.999, no_obj=2.271, cls=2.390
Epoch [29/50], Iter [350/418], Loss: total=1.907, reg=11.254,
containing_obj=6.972, no_obj=2.266, cls=2.397
Epoch [29/50], Iter [400/418], Loss: total=1.896, reg=11.189,
containing_obj=6.918, no_obj=2.266, cls=2.382
Time for epoch: 41.52 seconds
Updating best test loss: 2.58248

Starting epoch 30 / 50

Learning Rate for this epoch: 0.001

Epoch [30/50], Iter [50/418], Loss: total=1.826, reg=10.736,
containing_obj=6.708, no_obj=2.205, cls=2.258
Epoch [30/50], Iter [100/418], Loss: total=1.896, reg=11.206,
containing_obj=6.905, no_obj=2.264, cls=2.375
Epoch [30/50], Iter [150/418], Loss: total=1.918, reg=11.401,
containing_obj=6.947, no_obj=2.270, cls=2.397

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Epoch [30/50], Iter [200/418], Loss: total=1.915, reg=11.370,
containing_obj=6.959, no_obj=2.284, cls=2.365
Epoch [30/50], Iter [250/418], Loss: total=1.921, reg=11.437,
containing_obj=6.937, no_obj=2.282, cls=2.398
Epoch [30/50], Iter [300/418], Loss: total=1.912, reg=11.400,
containing_obj=6.909, no_obj=2.282, cls=2.353
Epoch [30/50], Iter [350/418], Loss: total=1.921, reg=11.433,
containing_obj=6.958, no_obj=2.261, cls=2.401
Epoch [30/50], Iter [400/418], Loss: total=1.916, reg=11.366,
containing_obj=6.950, no_obj=2.279, cls=2.396
Time for epoch: 68.58 seconds
---Evaluate model on test samples---

100%|      | 4950/4950 [02:00<00:00, 41.13it/s]

---class aeroplane ap 0.5263441952095804---
---class bicycle ap 0.6387820608269822---
---class bird ap 0.5081876621987862---
---class boat ap 0.30924617325664167---
---class bottle ap 0.1953919109939608---
---class bus ap 0.6054386390710843---
---class car ap 0.6859856258159108---
---class cat ap 0.6899663500308224---
---class chair ap 0.30562504518907185---
---class cow ap 0.5083984052418482---
---class diningtable ap 0.4030722397490757---
---class dog ap 0.6446285297768803---
---class horse ap 0.6609739749731018---
---class motorbike ap 0.5816842941969937---
---class person ap 0.5408731734147652---
---class pottedplant ap 0.16259329823464416---
---class sheep ap 0.48380072370711164---
---class sofa ap 0.4882684175741768---
---class train ap 0.6261446253901901---
---class tvmonitor ap 0.43276897704739115---
---map 0.499908716094951---
29 [np.float64(0.5263441952095804), np.float64(0.6387820608269822),
np.float64(0.5081876621987862), np.float64(0.30924617325664167),
np.float64(0.1953919109939608), np.float64(0.6054386390710843),
np.float64(0.6859856258159108), np.float64(0.6899663500308224),
np.float64(0.30562504518907185), np.float64(0.5083984052418482),
np.float64(0.4030722397490757), np.float64(0.6446285297768803),
np.float64(0.6609739749731018), np.float64(0.5816842941969937),
np.float64(0.5408731734147652), np.float64(0.16259329823464416),
np.float64(0.48380072370711164), np.float64(0.4882684175741768),
np.float64(0.6261446253901901), np.float64(0.43276897704739115)]
Updating best test loss: 2.56708

```

Starting epoch 31 / 50

Learning Rate for this epoch: 0.0001

Epoch [31/50], Iter [50/418], Loss: total=1.842, reg=10.817,
containing_obj=6.816, no_obj=2.255, cls=2.220
Epoch [31/50], Iter [100/418], Loss: total=1.819, reg=10.779,
containing_obj=6.749, no_obj=2.209, cls=2.098
Epoch [31/50], Iter [150/418], Loss: total=1.774, reg=10.538,
containing_obj=6.557, no_obj=2.216, cls=1.979
Epoch [31/50], Iter [200/418], Loss: total=1.814, reg=10.764,
containing_obj=6.781, no_obj=2.202, cls=2.019
Epoch [31/50], Iter [250/418], Loss: total=1.814, reg=10.790,
containing_obj=6.693, no_obj=2.215, cls=2.066
Epoch [31/50], Iter [300/418], Loss: total=1.802, reg=10.732,
containing_obj=6.662, no_obj=2.243, cls=1.990
Epoch [31/50], Iter [350/418], Loss: total=1.800, reg=10.736,
containing_obj=6.588, no_obj=2.251, cls=2.027
Epoch [31/50], Iter [400/418], Loss: total=1.802, reg=10.647,
containing_obj=6.645, no_obj=2.263, cls=2.071
Time for epoch: 72.81 seconds
Updating best test loss: 2.49617

Starting epoch 32 / 50

Learning Rate for this epoch: 0.0001

Epoch [32/50], Iter [50/418], Loss: total=1.797, reg=10.315,
containing_obj=6.882, no_obj=2.229, cls=2.140
Epoch [32/50], Iter [100/418], Loss: total=1.790, reg=10.345,
containing_obj=6.750, no_obj=2.267, cls=2.111
Epoch [32/50], Iter [150/418], Loss: total=1.775, reg=10.402,
containing_obj=6.653, no_obj=2.252, cls=1.998
Epoch [32/50], Iter [200/418], Loss: total=1.786, reg=10.458,
containing_obj=6.701, no_obj=2.244, cls=2.028
Epoch [32/50], Iter [250/418], Loss: total=1.795, reg=10.559,
containing_obj=6.734, no_obj=2.235, cls=2.014
Epoch [32/50], Iter [300/418], Loss: total=1.795, reg=10.570,
containing_obj=6.714, no_obj=2.226, cls=2.029
Epoch [32/50], Iter [350/418], Loss: total=1.780, reg=10.439,
containing_obj=6.674, no_obj=2.249, cls=1.998
Epoch [32/50], Iter [400/418], Loss: total=1.783, reg=10.440,
containing_obj=6.723, no_obj=2.245, cls=1.983
Time for epoch: 72.63 seconds
Updating best test loss: 2.49506

Starting epoch 33 / 50

Learning Rate for this epoch: 0.0001

Epoch [33/50], Iter [50/418], Loss: total=1.780, reg=10.342,
containing_obj=6.810, no_obj=2.311, cls=1.894

Epoch [33/50], Iter [100/418], Loss: total=1.728, reg=10.048,
containing_obj=6.476, no_obj=2.341, cls=1.874
Epoch [33/50], Iter [150/418], Loss: total=1.757, reg=10.266,
containing_obj=6.572, no_obj=2.273, cls=1.973
Epoch [33/50], Iter [200/418], Loss: total=1.763, reg=10.319,
containing_obj=6.591, no_obj=2.277, cls=1.968
Epoch [33/50], Iter [250/418], Loss: total=1.770, reg=10.362,
containing_obj=6.611, no_obj=2.273, cls=1.993
Epoch [33/50], Iter [300/418], Loss: total=1.774, reg=10.418,
containing_obj=6.608, no_obj=2.277, cls=1.987
Epoch [33/50], Iter [350/418], Loss: total=1.766, reg=10.374,
containing_obj=6.560, no_obj=2.268, cls=1.992
Epoch [33/50], Iter [400/418], Loss: total=1.754, reg=10.235,
containing_obj=6.572, no_obj=2.271, cls=1.974
Time for epoch: 56.68 seconds

Starting epoch 34 / 50

Learning Rate for this epoch: 0.0001

Epoch [34/50], Iter [50/418], Loss: total=1.831, reg=10.875,
containing_obj=7.012, no_obj=2.195, cls=1.886
Epoch [34/50], Iter [100/418], Loss: total=1.890, reg=11.210,
containing_obj=7.125, no_obj=2.228, cls=2.120
Epoch [34/50], Iter [150/418], Loss: total=1.815, reg=10.557,
containing_obj=6.887, no_obj=2.317, cls=2.018
Epoch [34/50], Iter [200/418], Loss: total=1.776, reg=10.291,
containing_obj=6.759, no_obj=2.336, cls=1.924
Epoch [34/50], Iter [250/418], Loss: total=1.776, reg=10.388,
containing_obj=6.694, no_obj=2.326, cls=1.902
Epoch [34/50], Iter [300/418], Loss: total=1.753, reg=10.264,
containing_obj=6.603, no_obj=2.313, cls=1.851
Epoch [34/50], Iter [350/418], Loss: total=1.752, reg=10.282,
containing_obj=6.629, no_obj=2.304, cls=1.815
Epoch [34/50], Iter [400/418], Loss: total=1.749, reg=10.230,
containing_obj=6.624, no_obj=2.293, cls=1.840
Time for epoch: 51.00 seconds
Updating best test loss: 2.49255

Starting epoch 35 / 50

Learning Rate for this epoch: 0.0001

Epoch [35/50], Iter [50/418], Loss: total=1.747, reg=10.219,
containing_obj=6.517, no_obj=2.235, cls=1.989
Epoch [35/50], Iter [100/418], Loss: total=1.701, reg=9.889,
containing_obj=6.308, no_obj=2.303, cls=1.913
Epoch [35/50], Iter [150/418], Loss: total=1.681, reg=9.726,
containing_obj=6.274, no_obj=2.300, cls=1.876
Epoch [35/50], Iter [200/418], Loss: total=1.679, reg=9.670,

```

containing_obj=6.290, no_obj=2.303, cls=1.888
Epoch [35/50], Iter [250/418], Loss: total=1.703, reg=9.838,
containing_obj=6.455, no_obj=2.278, cls=1.860
Epoch [35/50], Iter [300/418], Loss: total=1.707, reg=9.925,
containing_obj=6.431, no_obj=2.292, cls=1.832
Epoch [35/50], Iter [350/418], Loss: total=1.711, reg=9.995,
containing_obj=6.436, no_obj=2.281, cls=1.822
Epoch [35/50], Iter [400/418], Loss: total=1.720, reg=10.035,
containing_obj=6.476, no_obj=2.295, cls=1.833
Time for epoch: 71.86 seconds
---Evaluate model on test samples---

100%|      | 4950/4950 [01:57<00:00, 41.98it/s]

---class aeroplane ap 0.5245975914537888---
---class bicycle ap 0.6596501299188458---
---class bird ap 0.5242768765920766---
---class boat ap 0.3653802853059379---
---class bottle ap 0.20654025231705003---
---class bus ap 0.6158825840047565---
---class car ap 0.6795025855648448---
---class cat ap 0.7388607312260358---
---class chair ap 0.31881153955993874---
---class cow ap 0.537513772553205---
---class diningtable ap 0.3525117273131697---
---class dog ap 0.6835905887975844---
---class horse ap 0.6729047180219222---
---class motorbike ap 0.5907828569295864---
---class person ap 0.5460666980195533---
---class pottedplant ap 0.22079387862825636---
---class sheep ap 0.5013543221142717---
---class sofa ap 0.49176150890575365---
---class train ap 0.6436178000157954---
---class tvmonitor ap 0.4502612304562665---
---map 0.5162330838849319---
34 [np.float64(0.5245975914537888), np.float64(0.6596501299188458),
np.float64(0.5242768765920766), np.float64(0.3653802853059379),
np.float64(0.20654025231705003), np.float64(0.6158825840047565),
np.float64(0.6795025855648448), np.float64(0.7388607312260358),
np.float64(0.31881153955993874), np.float64(0.537513772553205),
np.float64(0.3525117273131697), np.float64(0.6835905887975844),
np.float64(0.6729047180219222), np.float64(0.5907828569295864),
np.float64(0.5460666980195533), np.float64(0.22079387862825636),
np.float64(0.5013543221142717), np.float64(0.49176150890575365),
np.float64(0.6436178000157954), np.float64(0.4502612304562665)]

```

```

Starting epoch 36 / 50
Learning Rate for this epoch: 0.0001

```

Epoch [36/50], Iter [50/418], Loss: total=1.757, reg=9.883,
containing_obj=6.688, no_obj=2.435, cls=2.073
Epoch [36/50], Iter [100/418], Loss: total=1.726, reg=9.699,
containing_obj=6.604, no_obj=2.372, cls=2.042
Epoch [36/50], Iter [150/418], Loss: total=1.769, reg=10.050,
containing_obj=6.817, no_obj=2.334, cls=2.031
Epoch [36/50], Iter [200/418], Loss: total=1.761, reg=10.079,
containing_obj=6.729, no_obj=2.317, cls=2.006
Epoch [36/50], Iter [250/418], Loss: total=1.744, reg=9.998,
containing_obj=6.575, no_obj=2.303, cls=2.049
Epoch [36/50], Iter [300/418], Loss: total=1.744, reg=10.065,
containing_obj=6.541, no_obj=2.302, cls=2.015
Epoch [36/50], Iter [350/418], Loss: total=1.732, reg=10.051,
containing_obj=6.497, no_obj=2.306, cls=1.928
Epoch [36/50], Iter [400/418], Loss: total=1.729, reg=9.978,
containing_obj=6.546, no_obj=2.297, cls=1.924
Time for epoch: 76.93 seconds
Updating best test loss: 2.49223

Starting epoch 37 / 50

Learning Rate for this epoch: 0.0001

Epoch [37/50], Iter [50/418], Loss: total=1.649, reg=9.715,
containing_obj=5.994, no_obj=2.298, cls=1.785
Epoch [37/50], Iter [100/418], Loss: total=1.692, reg=10.071,
containing_obj=6.235, no_obj=2.277, cls=1.718
Epoch [37/50], Iter [150/418], Loss: total=1.688, reg=9.924,
containing_obj=6.408, no_obj=2.256, cls=1.669
Epoch [37/50], Iter [200/418], Loss: total=1.703, reg=10.016,
containing_obj=6.434, no_obj=2.299, cls=1.690
Epoch [37/50], Iter [250/418], Loss: total=1.687, reg=9.941,
containing_obj=6.326, no_obj=2.322, cls=1.659
Epoch [37/50], Iter [300/418], Loss: total=1.698, reg=9.977,
containing_obj=6.360, no_obj=2.313, cls=1.720
Epoch [37/50], Iter [350/418], Loss: total=1.720, reg=10.149,
containing_obj=6.410, no_obj=2.326, cls=1.753
Epoch [37/50], Iter [400/418], Loss: total=1.726, reg=10.160,
containing_obj=6.442, no_obj=2.319, cls=1.787
Time for epoch: 76.31 seconds

Starting epoch 38 / 50

Learning Rate for this epoch: 0.0001

Epoch [38/50], Iter [50/418], Loss: total=1.643, reg=9.706,
containing_obj=5.955, no_obj=2.241, cls=1.815
Epoch [38/50], Iter [100/418], Loss: total=1.631, reg=9.478,
containing_obj=6.067, no_obj=2.284, cls=1.748
Epoch [38/50], Iter [150/418], Loss: total=1.633, reg=9.379,

containing_obj=6.143, no_obj=2.294, cls=1.780
Epoch [38/50], Iter [200/418], Loss: total=1.640, reg=9.430,
containing_obj=6.203, no_obj=2.302, cls=1.750
Epoch [38/50], Iter [250/418], Loss: total=1.672, reg=9.562,
containing_obj=6.430, no_obj=2.290, cls=1.783
Epoch [38/50], Iter [300/418], Loss: total=1.695, reg=9.739,
containing_obj=6.501, no_obj=2.300, cls=1.802
Epoch [38/50], Iter [350/418], Loss: total=1.696, reg=9.760,
containing_obj=6.444, no_obj=2.309, cls=1.840
Epoch [38/50], Iter [400/418], Loss: total=1.707, reg=9.858,
containing_obj=6.461, no_obj=2.325, cls=1.836
Time for epoch: 71.43 seconds
Updating best test loss: 2.46992

Starting epoch 39 / 50
Learning Rate for this epoch: 0.0001
Epoch [39/50], Iter [50/418], Loss: total=1.715, reg=10.067,
containing_obj=6.287, no_obj=2.363, cls=1.864
Epoch [39/50], Iter [100/418], Loss: total=1.703, reg=9.951,
containing_obj=6.239, no_obj=2.348, cls=1.897
Epoch [39/50], Iter [150/418], Loss: total=1.682, reg=9.897,
containing_obj=6.162, no_obj=2.327, cls=1.793
Epoch [39/50], Iter [200/418], Loss: total=1.664, reg=9.775,
containing_obj=6.173, no_obj=2.296, cls=1.722
Epoch [39/50], Iter [250/418], Loss: total=1.671, reg=9.799,
containing_obj=6.208, no_obj=2.304, cls=1.737
Epoch [39/50], Iter [300/418], Loss: total=1.683, reg=9.847,
containing_obj=6.265, no_obj=2.313, cls=1.773
Epoch [39/50], Iter [350/418], Loss: total=1.705, reg=9.984,
containing_obj=6.401, no_obj=2.297, cls=1.782
Epoch [39/50], Iter [400/418], Loss: total=1.708, reg=9.995,
containing_obj=6.424, no_obj=2.303, cls=1.769
Time for epoch: 51.54 seconds

Starting epoch 40 / 50
Learning Rate for this epoch: 0.0001
Epoch [40/50], Iter [50/418], Loss: total=1.667, reg=9.641,
containing_obj=6.088, no_obj=2.310, cls=1.963
Epoch [40/50], Iter [100/418], Loss: total=1.657, reg=9.560,
containing_obj=6.152, no_obj=2.369, cls=1.806
Epoch [40/50], Iter [150/418], Loss: total=1.716, reg=9.914,
containing_obj=6.378, no_obj=2.412, cls=1.890
Epoch [40/50], Iter [200/418], Loss: total=1.689, reg=9.684,
containing_obj=6.375, no_obj=2.359, cls=1.844
Epoch [40/50], Iter [250/418], Loss: total=1.698, reg=9.780,
containing_obj=6.369, no_obj=2.359, cls=1.866

```

Epoch [40/50], Iter [300/418], Loss: total=1.698, reg=9.777,
containing_obj=6.407, no_obj=2.355, cls=1.832
Epoch [40/50], Iter [350/418], Loss: total=1.687, reg=9.691,
containing_obj=6.384, no_obj=2.347, cls=1.824
Epoch [40/50], Iter [400/418], Loss: total=1.701, reg=9.749,
containing_obj=6.469, no_obj=2.332, cls=1.856
Time for epoch: 61.45 seconds
---Evaluate model on test samples---
100%|      | 4950/4950 [01:57<00:00, 42.15it/s]

---class aeroplane ap 0.506921372309586---
---class bicycle ap 0.6676564452767714---
---class bird ap 0.54628815452441---
---class boat ap 0.3374048783543677---
---class bottle ap 0.24039793126958686---
---class bus ap 0.5930549186632459---
---class car ap 0.6793849119827553---
---class cat ap 0.7337201518142095---
---class chair ap 0.3127342032391138---
---class cow ap 0.5216660925811383---
---class diningtable ap 0.3764727343579307---
---class dog ap 0.6958124188691409---
---class horse ap 0.678195154608118---
---class motorbike ap 0.569604056974025---
---class person ap 0.5572966331806313---
---class pottedplant ap 0.2613986135865137---
---class sheep ap 0.49829239992013724---
---class sofa ap 0.5151527068018484---
---class train ap 0.6337576999364929---
---class tvmonitor ap 0.4629447726926207---
---map 0.5194078125471322---
39 [np.float64(0.506921372309586), np.float64(0.6676564452767714),
np.float64(0.54628815452441), np.float64(0.3374048783543677),
np.float64(0.24039793126958686), np.float64(0.5930549186632459),
np.float64(0.6793849119827553), np.float64(0.7337201518142095),
np.float64(0.3127342032391138), np.float64(0.5216660925811383),
np.float64(0.3764727343579307), np.float64(0.6958124188691409),
np.float64(0.678195154608118), np.float64(0.569604056974025),
np.float64(0.5572966331806313), np.float64(0.2613986135865137),
np.float64(0.49829239992013724), np.float64(0.5151527068018484),
np.float64(0.6337576999364929), np.float64(0.4629447726926207)]

Starting epoch 41 / 50
Learning Rate for this epoch: 1e-05
Epoch [41/50], Iter [50/418], Loss: total=1.696, reg=9.920,
containing_obj=6.166, no_obj=2.454, cls=1.815
Epoch [41/50], Iter [100/418], Loss: total=1.662, reg=9.542,

```

containing_obj=6.106, no_obj=2.464, cls=1.828
Epoch [41/50], Iter [150/418], Loss: total=1.694, reg=9.691,
containing_obj=6.335, no_obj=2.435, cls=1.870
Epoch [41/50], Iter [200/418], Loss: total=1.715, reg=9.831,
containing_obj=6.488, no_obj=2.394, cls=1.864
Epoch [41/50], Iter [250/418], Loss: total=1.693, reg=9.756,
containing_obj=6.401, no_obj=2.374, cls=1.781
Epoch [41/50], Iter [300/418], Loss: total=1.685, reg=9.651,
containing_obj=6.392, no_obj=2.378, cls=1.796
Epoch [41/50], Iter [350/418], Loss: total=1.702, reg=9.813,
containing_obj=6.438, no_obj=2.377, cls=1.802
Epoch [41/50], Iter [400/418], Loss: total=1.701, reg=9.847,
containing_obj=6.411, no_obj=2.376, cls=1.783
Time for epoch: 77.22 seconds

Starting epoch 42 / 50

Learning Rate for this epoch: 1e-05

Epoch [42/50], Iter [50/418], Loss: total=1.830, reg=10.811,
containing_obj=6.965, no_obj=2.235, cls=1.950
Epoch [42/50], Iter [100/418], Loss: total=1.804, reg=10.656,
containing_obj=6.931, no_obj=2.230, cls=1.834
Epoch [42/50], Iter [150/418], Loss: total=1.760, reg=10.353,
containing_obj=6.708, no_obj=2.251, cls=1.813
Epoch [42/50], Iter [200/418], Loss: total=1.741, reg=10.174,
containing_obj=6.619, no_obj=2.291, cls=1.810
Epoch [42/50], Iter [250/418], Loss: total=1.710, reg=9.928,
containing_obj=6.475, no_obj=2.306, cls=1.807
Epoch [42/50], Iter [300/418], Loss: total=1.688, reg=9.798,
containing_obj=6.366, no_obj=2.308, cls=1.782
Epoch [42/50], Iter [350/418], Loss: total=1.687, reg=9.774,
containing_obj=6.376, no_obj=2.307, cls=1.789
Epoch [42/50], Iter [400/418], Loss: total=1.683, reg=9.737,
containing_obj=6.355, no_obj=2.320, cls=1.780
Time for epoch: 77.22 seconds

Starting epoch 43 / 50

Learning Rate for this epoch: 1e-05

Epoch [43/50], Iter [50/418], Loss: total=1.729, reg=10.001,
containing_obj=6.430, no_obj=2.395, cls=1.917
Epoch [43/50], Iter [100/418], Loss: total=1.722, reg=9.940,
containing_obj=6.516, no_obj=2.364, cls=1.845
Epoch [43/50], Iter [150/418], Loss: total=1.706, reg=9.859,
containing_obj=6.431, no_obj=2.345, cls=1.841
Epoch [43/50], Iter [200/418], Loss: total=1.710, reg=10.023,
containing_obj=6.397, no_obj=2.347, cls=1.758
Epoch [43/50], Iter [250/418], Loss: total=1.693, reg=9.932,

containing_obj=6.352, no_obj=2.336, cls=1.697
Epoch [43/50], Iter [300/418], Loss: total=1.726, reg=10.118,
containing_obj=6.489, no_obj=2.356, cls=1.747
Epoch [43/50], Iter [350/418], Loss: total=1.713, reg=9.992,
containing_obj=6.443, no_obj=2.358, cls=1.758
Epoch [43/50], Iter [400/418], Loss: total=1.705, reg=9.918,
containing_obj=6.462, no_obj=2.344, cls=1.739
Time for epoch: 76.89 seconds

Starting epoch 44 / 50

Learning Rate for this epoch: 1e-05

Epoch [44/50], Iter [50/418], Loss: total=1.612, reg=9.262,
containing_obj=6.129, no_obj=2.496, cls=1.459
Epoch [44/50], Iter [100/418], Loss: total=1.684, reg=9.937,
containing_obj=6.313, no_obj=2.361, cls=1.600
Epoch [44/50], Iter [150/418], Loss: total=1.705, reg=9.991,
containing_obj=6.399, no_obj=2.359, cls=1.716
Epoch [44/50], Iter [200/418], Loss: total=1.677, reg=9.825,
containing_obj=6.287, no_obj=2.368, cls=1.641
Epoch [44/50], Iter [250/418], Loss: total=1.685, reg=9.871,
containing_obj=6.365, no_obj=2.331, cls=1.654
Epoch [44/50], Iter [300/418], Loss: total=1.673, reg=9.727,
containing_obj=6.342, no_obj=2.332, cls=1.669
Epoch [44/50], Iter [350/418], Loss: total=1.684, reg=9.786,
containing_obj=6.368, no_obj=2.335, cls=1.714
Epoch [44/50], Iter [400/418], Loss: total=1.686, reg=9.802,
containing_obj=6.385, no_obj=2.328, cls=1.713
Time for epoch: 51.60 seconds
Updating best test loss: 2.46630

Starting epoch 45 / 50

Learning Rate for this epoch: 1e-05

Epoch [45/50], Iter [50/418], Loss: total=1.746, reg=10.191,
containing_obj=6.612, no_obj=2.297, cls=1.856
Epoch [45/50], Iter [100/418], Loss: total=1.702, reg=9.878,
containing_obj=6.498, no_obj=2.301, cls=1.748
Epoch [45/50], Iter [150/418], Loss: total=1.718, reg=10.032,
containing_obj=6.505, no_obj=2.316, cls=1.766
Epoch [45/50], Iter [200/418], Loss: total=1.713, reg=9.992,
containing_obj=6.470, no_obj=2.321, cls=1.771
Epoch [45/50], Iter [250/418], Loss: total=1.714, reg=9.983,
containing_obj=6.490, no_obj=2.322, cls=1.766
Epoch [45/50], Iter [300/418], Loss: total=1.699, reg=9.938,
containing_obj=6.394, no_obj=2.319, cls=1.736
Epoch [45/50], Iter [350/418], Loss: total=1.698, reg=9.912,
containing_obj=6.386, no_obj=2.337, cls=1.737

```

Epoch [45/50], Iter [400/418], Loss: total=1.701, reg=9.959,
containing_obj=6.389, no_obj=2.321, cls=1.740
Time for epoch: 54.74 seconds
---Evaluate model on test samples---
100%|      | 4950/4950 [01:57<00:00, 42.24it/s]

---class aeroplane ap 0.5261377250282667---
---class bicycle ap 0.6579281791560898---
---class bird ap 0.5410988351978443---
---class boat ap 0.3341373766897524---
---class bottle ap 0.22965435749231616---
---class bus ap 0.5757626389494326---
---class car ap 0.6767847273840715---
---class cat ap 0.7406366848980559---
---class chair ap 0.29660184994256034---
---class cow ap 0.5139298062605779---
---class diningtable ap 0.3587926506723184---
---class dog ap 0.6768755001010582---
---class horse ap 0.6638776885008171---
---class motorbike ap 0.5488583053954778---
---class person ap 0.544426920015931---
---class pottedplant ap 0.24625350435635113---
---class sheep ap 0.501016354339439---
---class sofa ap 0.5117036339736464---
---class train ap 0.6323670107623849---
---class tvmonitor ap 0.4474313458589143---
---map 0.5112137547487653---
44 [np.float64(0.5261377250282667), np.float64(0.6579281791560898),
np.float64(0.5410988351978443), np.float64(0.3341373766897524),
np.float64(0.22965435749231616), np.float64(0.5757626389494326),
np.float64(0.6767847273840715), np.float64(0.7406366848980559),
np.float64(0.29660184994256034), np.float64(0.5139298062605779),
np.float64(0.3587926506723184), np.float64(0.6768755001010582),
np.float64(0.6638776885008171), np.float64(0.5488583053954778),
np.float64(0.544426920015931), np.float64(0.24625350435635113),
np.float64(0.501016354339439), np.float64(0.5117036339736464),
np.float64(0.6323670107623849), np.float64(0.4474313458589143)]

```

Starting epoch 46 / 50

Learning Rate for this epoch: 1e-05

```

Epoch [46/50], Iter [50/418], Loss: total=1.716, reg=10.088,
containing_obj=6.290, no_obj=2.354, cls=1.865
Epoch [46/50], Iter [100/418], Loss: total=1.736, reg=10.059,
containing_obj=6.589, no_obj=2.334, cls=1.847
Epoch [46/50], Iter [150/418], Loss: total=1.692, reg=9.836,
containing_obj=6.330, no_obj=2.370, cls=1.773
Epoch [46/50], Iter [200/418], Loss: total=1.668, reg=9.712,

```

containing_obj=6.210, no_obj=2.352, cls=1.740
Epoch [46/50], Iter [250/418], Loss: total=1.687, reg=9.840,
containing_obj=6.328, no_obj=2.340, cls=1.737
Epoch [46/50], Iter [300/418], Loss: total=1.686, reg=9.788,
containing_obj=6.359, no_obj=2.342, cls=1.746
Epoch [46/50], Iter [350/418], Loss: total=1.687, reg=9.820,
containing_obj=6.336, no_obj=2.327, cls=1.761
Epoch [46/50], Iter [400/418], Loss: total=1.684, reg=9.795,
containing_obj=6.314, no_obj=2.332, cls=1.761
Time for epoch: 73.69 seconds

Starting epoch 47 / 50

Learning Rate for this epoch: 1e-05

Epoch [47/50], Iter [50/418], Loss: total=1.711, reg=9.783,
containing_obj=6.532, no_obj=2.249, cls=1.971
Epoch [47/50], Iter [100/418], Loss: total=1.721, reg=9.836,
containing_obj=6.584, no_obj=2.337, cls=1.899
Epoch [47/50], Iter [150/418], Loss: total=1.708, reg=9.866,
containing_obj=6.491, no_obj=2.341, cls=1.803
Epoch [47/50], Iter [200/418], Loss: total=1.682, reg=9.657,
containing_obj=6.449, no_obj=2.318, cls=1.756
Epoch [47/50], Iter [250/418], Loss: total=1.674, reg=9.544,
containing_obj=6.427, no_obj=2.331, cls=1.785
Epoch [47/50], Iter [300/418], Loss: total=1.688, reg=9.667,
containing_obj=6.467, no_obj=2.311, cls=1.814
Epoch [47/50], Iter [350/418], Loss: total=1.674, reg=9.581,
containing_obj=6.403, no_obj=2.331, cls=1.771
Epoch [47/50], Iter [400/418], Loss: total=1.672, reg=9.568,
containing_obj=6.374, no_obj=2.321, cls=1.796
Time for epoch: 74.23 seconds

Starting epoch 48 / 50

Learning Rate for this epoch: 1e-05

Epoch [48/50], Iter [50/418], Loss: total=1.595, reg=9.326,
containing_obj=6.002, no_obj=2.289, cls=1.519
Epoch [48/50], Iter [100/418], Loss: total=1.638, reg=9.389,
containing_obj=6.333, no_obj=2.372, cls=1.564
Epoch [48/50], Iter [150/418], Loss: total=1.634, reg=9.391,
containing_obj=6.319, no_obj=2.350, cls=1.547
Epoch [48/50], Iter [200/418], Loss: total=1.633, reg=9.330,
containing_obj=6.334, no_obj=2.343, cls=1.587
Epoch [48/50], Iter [250/418], Loss: total=1.634, reg=9.403,
containing_obj=6.286, no_obj=2.338, cls=1.577
Epoch [48/50], Iter [300/418], Loss: total=1.633, reg=9.429,
containing_obj=6.245, no_obj=2.329, cls=1.595
Epoch [48/50], Iter [350/418], Loss: total=1.646, reg=9.549,

containing_obj=6.292, no_obj=2.305, cls=1.604
Epoch [48/50], Iter [400/418], Loss: total=1.661, reg=9.601,
containing_obj=6.337, no_obj=2.283, cls=1.714
Time for epoch: 74.34 seconds

Starting epoch 49 / 50

Learning Rate for this epoch: 1e-05
Epoch [49/50], Iter [50/418], Loss: total=1.644, reg=9.629,
containing_obj=6.127, no_obj=2.238, cls=1.733
Epoch [49/50], Iter [100/418], Loss: total=1.680, reg=9.679,
containing_obj=6.344, no_obj=2.282, cls=1.856
Epoch [49/50], Iter [150/418], Loss: total=1.680, reg=9.814,
containing_obj=6.255, no_obj=2.291, cls=1.797
Epoch [49/50], Iter [200/418], Loss: total=1.664, reg=9.679,
containing_obj=6.263, no_obj=2.272, cls=1.757
Epoch [49/50], Iter [250/418], Loss: total=1.680, reg=9.799,
containing_obj=6.298, no_obj=2.291, cls=1.775
Epoch [49/50], Iter [300/418], Loss: total=1.698, reg=9.848,
containing_obj=6.387, no_obj=2.292, cls=1.846
Epoch [49/50], Iter [350/418], Loss: total=1.696, reg=9.876,
containing_obj=6.352, no_obj=2.293, cls=1.831
Epoch [49/50], Iter [400/418], Loss: total=1.687, reg=9.851,
containing_obj=6.300, no_obj=2.296, cls=1.798
Time for epoch: 52.19 seconds

Starting epoch 50 / 50

Learning Rate for this epoch: 1e-05
Epoch [50/50], Iter [50/418], Loss: total=1.727, reg=10.127,
containing_obj=6.490, no_obj=2.436, cls=1.673
Epoch [50/50], Iter [100/418], Loss: total=1.717, reg=10.115,
containing_obj=6.272, no_obj=2.440, cls=1.776
Epoch [50/50], Iter [150/418], Loss: total=1.713, reg=10.092,
containing_obj=6.297, no_obj=2.420, cls=1.751
Epoch [50/50], Iter [200/418], Loss: total=1.698, reg=9.953,
containing_obj=6.263, no_obj=2.395, cls=1.768
Epoch [50/50], Iter [250/418], Loss: total=1.724, reg=10.141,
containing_obj=6.366, no_obj=2.361, cls=1.819
Epoch [50/50], Iter [300/418], Loss: total=1.739, reg=10.251,
containing_obj=6.451, no_obj=2.331, cls=1.833
Epoch [50/50], Iter [350/418], Loss: total=1.722, reg=10.136,
containing_obj=6.440, no_obj=2.327, cls=1.765
Epoch [50/50], Iter [400/418], Loss: total=1.713, reg=10.087,
containing_obj=6.377, no_obj=2.312, cls=1.775
Time for epoch: 53.29 seconds
---Evaluate model on test samples---

```

100%|      | 4950/4950 [01:57<00:00, 42.05it/s]

---class aeroplane ap 0.5426723432364409---
---class bicycle ap 0.6715923245214914---
---class bird ap 0.5212549930744493---
---class boat ap 0.33196736198969246---
---class bottle ap 0.22175172937415366---
---class bus ap 0.6145423164477545---
---class car ap 0.6737025078008989---
---class cat ap 0.7474754055281722---
---class chair ap 0.32694559613610924---
---class cow ap 0.5105608880266215---
---class diningtable ap 0.40592310559961003---
---class dog ap 0.6800655758475856---
---class horse ap 0.6780774359847498---
---class motorbike ap 0.5711052256096341---
---class person ap 0.5418822853337416---
---class pottedplant ap 0.24695330596590503---
---class sheep ap 0.4972797878087374---
---class sofa ap 0.5126641815977018---
---class train ap 0.6287383464551739---
---class tvmonitor ap 0.45851247027063846---
---map 0.5191833593304631---
49 [np.float64(0.5426723432364409), np.float64(0.6715923245214914),
np.float64(0.5212549930744493), np.float64(0.33196736198969246),
np.float64(0.22175172937415366), np.float64(0.6145423164477545),
np.float64(0.6737025078008989), np.float64(0.7474754055281722),
np.float64(0.32694559613610924), np.float64(0.5105608880266215),
np.float64(0.40592310559961003), np.float64(0.6800655758475856),
np.float64(0.6780774359847498), np.float64(0.5711052256096341),
np.float64(0.5418822853337416), np.float64(0.24695330596590503),
np.float64(0.4972797878087374), np.float64(0.5126641815977018),
np.float64(0.6287383464551739), np.float64(0.45851247027063846)]

```

```

[20]: !curl -X POST -d "Training Done 152 Mi loss " \
      ntfy.sh/availableAstrak

```

```

{"id": "OnPbJ6qN8qFf", "time": 1745000760, "expires": 1745043960, "event": "message", "topic": "availableAstrak", "message": "Training Done 152 Mi loss "}

```

```

[ ]: if False:
    LATEST_NUMBER = 10

    # Load the network weights from epoch 5 checkpoint before training
    checkpoint_path = f'checkpoints/detector_epoch_{LATEST_NUMBER}.pth'
    if os.path.exists(checkpoint_path):
        print(f>Loading checkpoint from {checkpoint_path})

```

```

        net.load_state_dict(torch.load(checkpoint_path, map_location=device))
    else:
        print(f"Checkpoint {checkpoint_path} not found. Training from scratch.")

    best_test_loss = np.inf
    learning_rate = learning_rate

    os.makedirs('./checkpoints', exist_ok=True)
    for epoch in range(LATEST_NUMBER, num_epochs):
        net.train()

        # Update learning rate late in training
        if epoch == 30 or epoch == 40:
            learning_rate /= 10.0

        for param_group in optimizer.param_groups:
            param_group['lr'] = learning_rate

        print('\n\nStarting epoch %d / %d' % (epoch + 1, num_epochs))
        print('Learning Rate for this epoch: {}'.format(learning_rate))

        total_loss = collections.defaultdict(int)

        start_iteration_time = time.time()
        for i, data in enumerate(train_loader):
            data = (item.to(device) for item in data)
            images, target_boxes, target_cls, has_object_map = data
            pred = net(images)
            loss_dict = criterion(pred, target_boxes, target_cls,
↪has_object_map)
            for key in loss_dict:
                total_loss[key] += loss_dict[key].item()

            optimizer.zero_grad()
            loss_dict['total_loss'].backward()
            optimizer.step()

            if (i+1) % 50 == 0:
                outstring = 'Epoch [%d/%d], Iter [%d/%d], Loss: ' % ((epoch+1,
↪num_epochs, i+1, len(train_loader)))
                outstring += ', '.join( "%s=%.3f" % (key[:5], val / (i+1)) for
↪key, val in total_loss.items() )
                print(outstring)
            print('Time for epoch: %.2f seconds' % (time.time() -
↪start_iteration_time))

        # evaluate the network on the test data

```

```

        if (epoch + 1) % 5 == 0:
            test_aps = evaluate(net, test_dataset_file=annotation_file_test,
img_root=file_root_test)
            print(epoch, test_aps)
            with torch.no_grad():
                test_loss = 0.0
                net.eval()
                for i, data in enumerate(test_loader):
                    data = (item.to(device) for item in data)
                    images, target_boxes, target_cls, has_object_map = data

                    pred = net(images)
                    loss_dict = criterion(pred, target_boxes, target_cls,
img_has_object_map)
                    test_loss += loss_dict['total_loss'].item()
                test_loss /= len(test_loader)

            if best_test_loss > test_loss:
                best_test_loss = test_loss
                print('Updating best test loss: %.5f' % best_test_loss)
                torch.save(net.state_dict(), 'checkpoints/best_detector.pth')

            if (epoch+1) in [5, 10, 15, 20, 25, 30, 35, 40]:
                torch.save(net.state_dict(), 'checkpoints/detector_epoch_%d.pth' %
img(epoch+1))

            torch.save(net.state_dict(), 'checkpoints/detector.pth')

```

Loading checkpoint from checkpoints/detector_epoch_10.pth

Starting epoch 11 / 50

Learning Rate for this epoch: 0.001

```

Epoch [11/50], Iter [50/418], Loss: total=3.626, reg=19.548,
containing_obj=9.190, no_obj=1.081, cls=13.697
Epoch [11/50], Iter [100/418], Loss: total=3.700, reg=19.715,
containing_obj=9.444, no_obj=1.091, cls=14.149
Epoch [11/50], Iter [150/418], Loss: total=3.710, reg=19.986,
containing_obj=9.362, no_obj=1.105, cls=14.063
Epoch [11/50], Iter [200/418], Loss: total=3.683, reg=19.760,
containing_obj=9.313, no_obj=1.102, cls=14.016
Epoch [11/50], Iter [250/418], Loss: total=3.694, reg=19.911,
containing_obj=9.310, no_obj=1.112, cls=13.996
Epoch [11/50], Iter [300/418], Loss: total=3.677, reg=19.910,
containing_obj=9.313, no_obj=1.107, cls=13.787
Epoch [11/50], Iter [350/418], Loss: total=3.697, reg=19.946,
containing_obj=9.375, no_obj=1.111, cls=13.931

```

Epoch [11/50], Iter [400/418], Loss: total=3.674, reg=19.827,
containing_obj=9.348, no_obj=1.113, cls=13.797
Time for epoch: 200.73 seconds
Updating best test loss: 3.73059

Starting epoch 12 / 50

Learning Rate for this epoch: 0.001

Epoch [12/50], Iter [50/418], Loss: total=3.551, reg=19.083,
containing_obj=9.425, no_obj=1.168, cls=12.940
Epoch [12/50], Iter [100/418], Loss: total=3.560, reg=19.350,
containing_obj=9.098, no_obj=1.155, cls=13.112
Epoch [12/50], Iter [150/418], Loss: total=3.539, reg=19.266,
containing_obj=9.086, no_obj=1.158, cls=12.961
Epoch [12/50], Iter [200/418], Loss: total=3.563, reg=19.254,
containing_obj=9.169, no_obj=1.177, cls=13.150
Epoch [12/50], Iter [250/418], Loss: total=3.546, reg=19.229,
containing_obj=9.179, no_obj=1.178, cls=12.969
Epoch [12/50], Iter [300/418], Loss: total=3.562, reg=19.272,
containing_obj=9.235, no_obj=1.182, cls=13.051
Epoch [12/50], Iter [350/418], Loss: total=3.547, reg=19.198,
containing_obj=9.230, no_obj=1.192, cls=12.948
Epoch [12/50], Iter [400/418], Loss: total=3.565, reg=19.332,
containing_obj=9.313, no_obj=1.192, cls=12.948
Time for epoch: 194.92 seconds
Updating best test loss: 3.62728

Starting epoch 13 / 50

Learning Rate for this epoch: 0.001

Epoch [13/50], Iter [50/418], Loss: total=3.568, reg=19.845,
containing_obj=9.662, no_obj=1.332, cls=11.981
Epoch [13/50], Iter [100/418], Loss: total=3.508, reg=19.500,
containing_obj=9.313, no_obj=1.325, cls=11.962
Epoch [13/50], Iter [150/418], Loss: total=3.567, reg=19.796,
containing_obj=9.455, no_obj=1.300, cls=12.253
Epoch [13/50], Iter [200/418], Loss: total=3.543, reg=19.600,
containing_obj=9.354, no_obj=1.313, cls=12.247
Epoch [13/50], Iter [250/418], Loss: total=3.519, reg=19.609,
containing_obj=9.305, no_obj=1.318, cls=11.995
Epoch [13/50], Iter [300/418], Loss: total=3.481, reg=19.381,
containing_obj=9.271, no_obj=1.318, cls=11.799
Epoch [13/50], Iter [350/418], Loss: total=3.466, reg=19.266,
containing_obj=9.258, no_obj=1.321, cls=11.748
Epoch [13/50], Iter [400/418], Loss: total=3.447, reg=19.082,
containing_obj=9.272, no_obj=1.324, cls=11.689
Time for epoch: 190.10 seconds
Updating best test loss: 3.50319

Starting epoch 14 / 50

Learning Rate for this epoch: 0.001

Epoch [14/50], Iter [50/418], Loss: total=3.344, reg=19.157,
containing_obj=9.340, no_obj=1.355, cls=10.276
Epoch [14/50], Iter [100/418], Loss: total=3.332, reg=18.816,
containing_obj=9.129, no_obj=1.409, cls=10.630
Epoch [14/50], Iter [150/418], Loss: total=3.249, reg=18.175,
containing_obj=9.003, no_obj=1.425, cls=10.384
Epoch [14/50], Iter [200/418], Loss: total=3.269, reg=18.331,
containing_obj=9.056, no_obj=1.432, cls=10.408
Epoch [14/50], Iter [250/418], Loss: total=3.291, reg=18.427,
containing_obj=9.164, no_obj=1.430, cls=10.475
Epoch [14/50], Iter [300/418], Loss: total=3.293, reg=18.452,
containing_obj=9.139, no_obj=1.433, cls=10.486
Epoch [14/50], Iter [350/418], Loss: total=3.272, reg=18.344,
containing_obj=9.101, no_obj=1.436, cls=10.379
Epoch [14/50], Iter [400/418], Loss: total=3.254, reg=18.304,
containing_obj=9.070, no_obj=1.430, cls=10.249
Time for epoch: 190.16 seconds
Updating best test loss: 3.35076

Starting epoch 15 / 50

Learning Rate for this epoch: 0.001

Epoch [15/50], Iter [50/418], Loss: total=3.216, reg=18.363,
containing_obj=9.055, no_obj=1.522, cls=9.655
Epoch [15/50], Iter [100/418], Loss: total=3.165, reg=18.126,
containing_obj=8.971, no_obj=1.484, cls=9.397
Epoch [15/50], Iter [150/418], Loss: total=3.110, reg=17.772,
containing_obj=8.884, no_obj=1.453, cls=9.214
Epoch [15/50], Iter [200/418], Loss: total=3.127, reg=17.791,
containing_obj=9.005, no_obj=1.430, cls=9.303
Epoch [15/50], Iter [250/418], Loss: total=3.141, reg=17.818,
containing_obj=9.132, no_obj=1.438, cls=9.300
Epoch [15/50], Iter [300/418], Loss: total=3.160, reg=18.083,
containing_obj=9.153, no_obj=1.454, cls=9.236
Epoch [15/50], Iter [350/418], Loss: total=3.155, reg=18.068,
containing_obj=9.080, no_obj=1.469, cls=9.242
Epoch [15/50], Iter [400/418], Loss: total=3.135, reg=17.925,
containing_obj=9.034, no_obj=1.466, cls=9.196
Time for epoch: 332.56 seconds
---Evaluate model on test samples---
100%| | 4950/4950 [01:24<00:00, 58.49it/s]
---class aeroplane ap 0.34016033181303---
---class bicycle ap 0.4065908724442966---

```

---class bird ap 0.19900911722002484---
---class boat ap 0.054747304825003625---
---class bottle ap 0.027618340528911888---
---class bus ap 0.11905099359258275---
---class car ap 0.4573312886031934---
---class cat ap 0.3836367591198119---
---class chair ap 0.14139145900766487---
---class cow ap 0.17038890955455666---
---class diningtable ap 0.14481027130586824---
---class dog ap 0.4364369941505462---
---class horse ap 0.4706343658270318---
---class motorbike ap 0.39791972750975113---
---class person ap 0.36227004168001953---
---class pottedplant ap 0.04054494191505592---
---class sheep ap 0.1629491016382147---
---class sofa ap 0.0--- (no predictions for this class)
---class train ap 0.3817067627538299---
---class tvmonitor ap 0.14641063548434136---
---map 0.24218041094868675---
14 [np.float64(0.34016033181303), np.float64(0.4065908724442966),
np.float64(0.19900911722002484), np.float64(0.054747304825003625),
np.float64(0.027618340528911888), np.float64(0.11905099359258275),
np.float64(0.4573312886031934), np.float64(0.3836367591198119),
np.float64(0.14139145900766487), np.float64(0.17038890955455666),
np.float64(0.14481027130586824), np.float64(0.4364369941505462),
np.float64(0.4706343658270318), np.float64(0.39791972750975113),
np.float64(0.36227004168001953), np.float64(0.04054494191505592),
np.float64(0.1629491016382147), 0.0, np.float64(0.3817067627538299),
np.float64(0.14641063548434136)]
Updating best test loss: 3.22515

```

Starting epoch 16 / 50

Learning Rate for this epoch: 0.001

```

Epoch [16/50], Iter [50/418], Loss: total=3.096, reg=17.964,
containing_obj=8.846, no_obj=1.572, cls=8.770
Epoch [16/50], Iter [100/418], Loss: total=3.045, reg=17.729,
containing_obj=8.775, no_obj=1.566, cls=8.474
Epoch [16/50], Iter [150/418], Loss: total=3.002, reg=17.457,
containing_obj=8.750, no_obj=1.553, cls=8.265
Epoch [16/50], Iter [200/418], Loss: total=3.010, reg=17.466,
containing_obj=8.771, no_obj=1.571, cls=8.316
Epoch [16/50], Iter [250/418], Loss: total=3.014, reg=17.462,
containing_obj=8.791, no_obj=1.576, cls=8.342
Epoch [16/50], Iter [300/418], Loss: total=3.033, reg=17.560,
containing_obj=8.816, no_obj=1.565, cls=8.455
Epoch [16/50], Iter [350/418], Loss: total=3.042, reg=17.599,
containing_obj=8.835, no_obj=1.574, cls=8.494

```

Epoch [16/50], Iter [400/418], Loss: total=3.067, reg=17.737,
containing_obj=8.933, no_obj=1.570, cls=8.567
Time for epoch: 212.87 seconds
Updating best test loss: 3.12361

Starting epoch 17 / 50

Learning Rate for this epoch: 0.001

Epoch [17/50], Iter [50/418], Loss: total=3.042, reg=17.823,
containing_obj=9.554, no_obj=1.551, cls=7.571
Epoch [17/50], Iter [100/418], Loss: total=2.951, reg=17.017,
containing_obj=9.155, no_obj=1.592, cls=7.646
Epoch [17/50], Iter [150/418], Loss: total=2.974, reg=17.120,
containing_obj=9.087, no_obj=1.566, cls=7.917
Epoch [17/50], Iter [200/418], Loss: total=2.949, reg=17.003,
containing_obj=8.938, no_obj=1.581, cls=7.863
Epoch [17/50], Iter [250/418], Loss: total=2.967, reg=17.230,
containing_obj=9.023, no_obj=1.582, cls=7.774
Epoch [17/50], Iter [300/418], Loss: total=2.971, reg=17.251,
containing_obj=8.970, no_obj=1.605, cls=7.825
Epoch [17/50], Iter [350/418], Loss: total=2.971, reg=17.275,
containing_obj=8.925, no_obj=1.605, cls=7.842
Epoch [17/50], Iter [400/418], Loss: total=2.985, reg=17.372,
containing_obj=8.924, no_obj=1.618, cls=7.904
Time for epoch: 1450.94 seconds
Updating best test loss: 3.05840

Starting epoch 18 / 50

Learning Rate for this epoch: 0.001

Epoch [18/50], Iter [50/418], Loss: total=2.790, reg=16.451,
containing_obj=8.579, no_obj=1.693, cls=6.753
Epoch [18/50], Iter [100/418], Loss: total=2.771, reg=16.279,
containing_obj=8.641, no_obj=1.644, cls=6.691
Epoch [18/50], Iter [150/418], Loss: total=2.877, reg=16.816,
containing_obj=8.909, no_obj=1.637, cls=7.159
Epoch [18/50], Iter [200/418], Loss: total=2.895, reg=16.871,
containing_obj=8.954, no_obj=1.648, cls=7.270
Epoch [18/50], Iter [250/418], Loss: total=2.883, reg=16.786,
containing_obj=8.919, no_obj=1.640, cls=7.250
Epoch [18/50], Iter [300/418], Loss: total=2.885, reg=16.805,
containing_obj=8.913, no_obj=1.651, cls=7.247
Epoch [18/50], Iter [350/418], Loss: total=2.900, reg=16.970,
containing_obj=8.917, no_obj=1.666, cls=7.242
Epoch [18/50], Iter [400/418], Loss: total=2.907, reg=17.031,
containing_obj=8.945, no_obj=1.667, cls=7.246
Time for epoch: 571.32 seconds
Updating best test loss: 3.02741

Starting epoch 19 / 50

Learning Rate for this epoch: 0.001

Epoch [19/50], Iter [50/418], Loss: total=2.846, reg=16.114,
containing_obj=9.123, no_obj=1.686, cls=7.233
Epoch [19/50], Iter [100/418], Loss: total=2.898, reg=16.932,
containing_obj=8.899, no_obj=1.702, cls=7.240
Epoch [19/50], Iter [150/418], Loss: total=2.893, reg=16.961,
containing_obj=8.800, no_obj=1.704, cls=7.255
Epoch [19/50], Iter [200/418], Loss: total=2.896, reg=17.033,
containing_obj=8.877, no_obj=1.712, cls=7.130
Epoch [19/50], Iter [250/418], Loss: total=2.918, reg=17.197,
containing_obj=8.913, no_obj=1.739, cls=7.168
Epoch [19/50], Iter [300/418], Loss: total=2.877, reg=16.904,
containing_obj=8.814, no_obj=1.758, cls=7.047
Epoch [19/50], Iter [350/418], Loss: total=2.846, reg=16.714,
containing_obj=8.775, no_obj=1.743, cls=6.921
Epoch [19/50], Iter [400/418], Loss: total=2.845, reg=16.742,
containing_obj=8.769, no_obj=1.739, cls=6.886
Time for epoch: 2576.50 seconds
Updating best test loss: 2.96833

Starting epoch 20 / 50

Learning Rate for this epoch: 0.001

Epoch [20/50], Iter [50/418], Loss: total=2.691, reg=15.563,
containing_obj=8.403, no_obj=1.744, cls=6.588
Epoch [20/50], Iter [100/418], Loss: total=2.705, reg=15.771,
containing_obj=8.686, no_obj=1.740, cls=6.267
Epoch [20/50], Iter [150/418], Loss: total=2.726, reg=15.950,
containing_obj=8.680, no_obj=1.757, cls=6.323
Epoch [20/50], Iter [200/418], Loss: total=2.721, reg=15.952,
containing_obj=8.675, no_obj=1.754, cls=6.272
Epoch [20/50], Iter [250/418], Loss: total=2.714, reg=15.891,
containing_obj=8.685, no_obj=1.750, cls=6.237
Epoch [20/50], Iter [300/418], Loss: total=2.728, reg=15.980,
containing_obj=8.733, no_obj=1.773, cls=6.252
Epoch [20/50], Iter [350/418], Loss: total=2.731, reg=16.063,
containing_obj=8.706, no_obj=1.786, cls=6.218
Epoch [20/50], Iter [400/418], Loss: total=2.724, reg=16.056,
containing_obj=8.688, no_obj=1.781, cls=6.164
Time for epoch: 923.94 seconds
---Evaluate model on test samples---
100%| | 4950/4950 [02:14<00:00, 36.79it/s]
---class aeroplane ap 0.38472875723928734---
---class bicycle ap 0.39253685234853386---

```

---class bird ap 0.2924927166093988---
---class boat ap 0.12330638744535125---
---class bottle ap 0.05951869568500942---
---class bus ap 0.4355312338470818---
---class car ap 0.48431811638504807---
---class cat ap 0.5322518906615528---
---class chair ap 0.24382562176691583---
---class cow ap 0.2856163239314491---
---class diningtable ap 0.2681859152200399---
---class dog ap 0.4867678876139076---
---class horse ap 0.5496349341368674---
---class motorbike ap 0.36943403003124176---
---class person ap 0.32209095171614577---
---class pottedplant ap 0.09648933683146042---
---class sheep ap 0.32550334964903616---
---class sofa ap 0.23077582296411175---
---class train ap 0.44403955507537207---
---class tvmonitor ap 0.30759965709227605---
---map 0.3317324018125044---
19 [np.float64(0.38472875723928734), np.float64(0.39253685234853386),
np.float64(0.2924927166093988), np.float64(0.12330638744535125),
np.float64(0.05951869568500942), np.float64(0.4355312338470818),
np.float64(0.48431811638504807), np.float64(0.5322518906615528),
np.float64(0.24382562176691583), np.float64(0.2856163239314491),
np.float64(0.2681859152200399), np.float64(0.4867678876139076),
np.float64(0.5496349341368674), np.float64(0.36943403003124176),
np.float64(0.32209095171614577), np.float64(0.09648933683146042),
np.float64(0.32550334964903616), np.float64(0.23077582296411175),
np.float64(0.44403955507537207), np.float64(0.30759965709227605)]
Updating best test loss: 2.90832

```

Starting epoch 21 / 50

Learning Rate for this epoch: 0.001

```

Epoch [21/50], Iter [50/418], Loss: total=2.555, reg=14.812,
containing_obj=8.507, no_obj=1.713, cls=5.623
Epoch [21/50], Iter [100/418], Loss: total=2.623, reg=15.510,
containing_obj=8.811, no_obj=1.723, cls=5.436
Epoch [21/50], Iter [150/418], Loss: total=2.603, reg=15.392,
containing_obj=8.570, no_obj=1.769, cls=5.498
Epoch [21/50], Iter [200/418], Loss: total=2.641, reg=15.598,
containing_obj=8.651, no_obj=1.788, cls=5.652
Epoch [21/50], Iter [250/418], Loss: total=2.673, reg=15.783,
containing_obj=8.675, no_obj=1.789, cls=5.830
Epoch [21/50], Iter [300/418], Loss: total=2.647, reg=15.602,
containing_obj=8.595, no_obj=1.783, cls=5.780
Epoch [21/50], Iter [350/418], Loss: total=2.653, reg=15.677,
containing_obj=8.592, no_obj=1.785, cls=5.779

```

Epoch [21/50], Iter [400/418], Loss: total=2.669, reg=15.714,
containing_obj=8.690, no_obj=1.780, cls=5.842
Time for epoch: 292.94 seconds
Updating best test loss: 2.86743

Starting epoch 22 / 50

Learning Rate for this epoch: 0.001

Epoch [22/50], Iter [50/418], Loss: total=2.639, reg=15.536,
containing_obj=8.937, no_obj=1.767, cls=5.434
Epoch [22/50], Iter [100/418], Loss: total=2.723, reg=16.107,
containing_obj=8.884, no_obj=1.828, cls=5.857
Epoch [22/50], Iter [150/418], Loss: total=2.671, reg=15.749,
containing_obj=8.668, no_obj=1.817, cls=5.816
Epoch [22/50], Iter [200/418], Loss: total=2.702, reg=15.965,
containing_obj=8.776, no_obj=1.824, cls=5.859
Epoch [22/50], Iter [250/418], Loss: total=2.685, reg=15.833,
containing_obj=8.750, no_obj=1.837, cls=5.806
Epoch [22/50], Iter [300/418], Loss: total=2.682, reg=15.843,
containing_obj=8.746, no_obj=1.842, cls=5.752
Epoch [22/50], Iter [350/418], Loss: total=2.665, reg=15.689,
containing_obj=8.681, no_obj=1.863, cls=5.743
Epoch [22/50], Iter [400/418], Loss: total=2.641, reg=15.560,
containing_obj=8.610, no_obj=1.866, cls=5.651
Time for epoch: 3751.61 seconds
Updating best test loss: 2.85673

Starting epoch 23 / 50

Learning Rate for this epoch: 0.001

Epoch [23/50], Iter [50/418], Loss: total=2.620, reg=15.498,
containing_obj=8.545, no_obj=1.895, cls=5.498
Epoch [23/50], Iter [100/418], Loss: total=2.627, reg=15.810,
containing_obj=8.693, no_obj=1.886, cls=5.130
Epoch [23/50], Iter [150/418], Loss: total=2.559, reg=15.246,
containing_obj=8.518, no_obj=1.880, cls=5.070
Epoch [23/50], Iter [200/418], Loss: total=2.577, reg=15.348,
containing_obj=8.515, no_obj=1.905, cls=5.159
Epoch [23/50], Iter [250/418], Loss: total=2.575, reg=15.359,
containing_obj=8.491, no_obj=1.916, cls=5.129
Epoch [23/50], Iter [300/418], Loss: total=2.591, reg=15.526,
containing_obj=8.475, no_obj=1.923, cls=5.172
Epoch [23/50], Iter [350/418], Loss: total=2.576, reg=15.479,
containing_obj=8.433, no_obj=1.919, cls=5.084
Epoch [23/50], Iter [400/418], Loss: total=2.599, reg=15.564,
containing_obj=8.482, no_obj=1.921, cls=5.218
Time for epoch: 1271.38 seconds
Updating best test loss: 2.80972

Starting epoch 24 / 50
 Learning Rate for this epoch: 0.001
 Epoch [24/50], Iter [50/418], Loss: total=2.519, reg=14.747,
 containing_obj=8.747, no_obj=1.931, cls=4.800
 Epoch [24/50], Iter [100/418], Loss: total=2.486, reg=14.761,
 containing_obj=8.292, no_obj=1.986, cls=4.794
 Epoch [24/50], Iter [150/418], Loss: total=2.476, reg=14.545,
 containing_obj=8.367, no_obj=1.981, cls=4.814
 Epoch [24/50], Iter [200/418], Loss: total=2.449, reg=14.409,
 containing_obj=8.313, no_obj=1.983, cls=4.681
 Epoch [24/50], Iter [250/418], Loss: total=2.457, reg=14.480,
 containing_obj=8.332, no_obj=1.983, cls=4.691
 Epoch [24/50], Iter [300/418], Loss: total=2.484, reg=14.666,
 containing_obj=8.371, no_obj=1.955, cls=4.817
 Epoch [24/50], Iter [350/418], Loss: total=2.512, reg=14.917,
 containing_obj=8.426, no_obj=1.947, cls=4.858
 Epoch [24/50], Iter [400/418], Loss: total=2.518, reg=15.001,
 containing_obj=8.443, no_obj=1.961, cls=4.815
 Time for epoch: 346.54 seconds
 Updating best test loss: 2.78412

Starting epoch 25 / 50
 Learning Rate for this epoch: 0.001
 Epoch [25/50], Iter [50/418], Loss: total=2.526, reg=15.035,
 containing_obj=8.460, no_obj=1.994, cls=4.829
 Epoch [25/50], Iter [100/418], Loss: total=2.451, reg=14.426,
 containing_obj=8.317, no_obj=1.985, cls=4.687
 Epoch [25/50], Iter [150/418], Loss: total=2.485, reg=14.752,
 containing_obj=8.323, no_obj=1.972, cls=4.773
 Epoch [25/50], Iter [200/418], Loss: total=2.468, reg=14.517,
 containing_obj=8.327, no_obj=2.003, cls=4.770
 Epoch [25/50], Iter [250/418], Loss: total=2.465, reg=14.472,
 containing_obj=8.312, no_obj=2.015, cls=4.785
 Epoch [25/50], Iter [300/418], Loss: total=2.476, reg=14.575,
 containing_obj=8.399, no_obj=1.996, cls=4.737
 Epoch [25/50], Iter [350/418], Loss: total=2.476, reg=14.586,
 containing_obj=8.421, no_obj=1.997, cls=4.708
 Epoch [25/50], Iter [400/418], Loss: total=2.475, reg=14.614,
 containing_obj=8.375, no_obj=1.993, cls=4.722
 Time for epoch: 347.48 seconds
 ---Evaluate model on test samples---
 100%| | 4950/4950 [03:07<00:00, 26.43it/s]
 ---class aeroplane ap 0.40535871694803755---
 ---class bicycle ap 0.5373190564347323---

```

---class bird ap 0.3820299267072676---
---class boat ap 0.23216683241051325---
---class bottle ap 0.10816627053971814---
---class bus ap 0.5342607062473571---
---class car ap 0.513542351026582---
---class cat ap 0.6429760184608465---
---class chair ap 0.2308024504718296---
---class cow ap 0.42249877949789827---
---class diningtable ap 0.25376710756600906---
---class dog ap 0.5332279772185425---
---class horse ap 0.5730518512406653---
---class motorbike ap 0.4753764101256551---
---class person ap 0.39084660985014164---
---class pottedplant ap 0.0925682583412986---
---class sheep ap 0.3711291445306785---
---class sofa ap 0.3650761387832203---
---class train ap 0.5767696606798006---
---class tvmonitor ap 0.37291497491275283---
---map 0.40069246209967735---
24 [np.float64(0.40535871694803755), np.float64(0.5373190564347323),
np.float64(0.3820299267072676), np.float64(0.23216683241051325),
np.float64(0.10816627053971814), np.float64(0.5342607062473571),
np.float64(0.513542351026582), np.float64(0.6429760184608465),
np.float64(0.2308024504718296), np.float64(0.42249877949789827),
np.float64(0.25376710756600906), np.float64(0.5332279772185425),
np.float64(0.5730518512406653), np.float64(0.4753764101256551),
np.float64(0.39084660985014164), np.float64(0.0925682583412986),
np.float64(0.3711291445306785), np.float64(0.3650761387832203),
np.float64(0.5767696606798006), np.float64(0.37291497491275283)]
Updating best test loss: 2.75176

```

Starting epoch 26 / 50

Learning Rate for this epoch: 0.001

```

Epoch [26/50], Iter [50/418], Loss: total=2.576, reg=15.536,
containing_obj=8.392, no_obj=2.005, cls=4.974
Epoch [26/50], Iter [100/418], Loss: total=2.526, reg=15.394,
containing_obj=8.383, no_obj=1.955, cls=4.578
Epoch [26/50], Iter [150/418], Loss: total=2.483, reg=14.913,
containing_obj=8.231, no_obj=1.998, cls=4.655
Epoch [26/50], Iter [200/418], Loss: total=2.479, reg=14.832,
containing_obj=8.320, no_obj=2.040, cls=4.552
Epoch [26/50], Iter [250/418], Loss: total=2.458, reg=14.727,
containing_obj=8.220, no_obj=2.051, cls=4.497
Epoch [26/50], Iter [300/418], Loss: total=2.444, reg=14.629,
containing_obj=8.192, no_obj=2.054, cls=4.456
Epoch [26/50], Iter [350/418], Loss: total=2.436, reg=14.540,
containing_obj=8.229, no_obj=2.043, cls=4.414

```

Epoch [26/50], Iter [400/418], Loss: total=2.446, reg=14.625,
containing_obj=8.239, no_obj=2.042, cls=4.445
Time for epoch: 348.79 seconds

Starting epoch 27 / 50

Learning Rate for this epoch: 0.001

Epoch [27/50], Iter [50/418], Loss: total=2.156, reg=12.445,
containing_obj=7.792, no_obj=1.949, cls=3.691
Epoch [27/50], Iter [100/418], Loss: total=2.249, reg=13.137,
containing_obj=7.909, no_obj=1.969, cls=3.974
Epoch [27/50], Iter [150/418], Loss: total=2.344, reg=13.751,
containing_obj=8.141, no_obj=1.984, cls=4.250
Epoch [27/50], Iter [200/418], Loss: total=2.365, reg=13.896,
containing_obj=8.169, no_obj=2.005, cls=4.310
Epoch [27/50], Iter [250/418], Loss: total=2.387, reg=14.090,
containing_obj=8.223, no_obj=2.009, cls=4.322
Epoch [27/50], Iter [300/418], Loss: total=2.401, reg=14.177,
containing_obj=8.338, no_obj=2.039, cls=4.263
Epoch [27/50], Iter [350/418], Loss: total=2.374, reg=13.992,
containing_obj=8.245, no_obj=2.044, cls=4.212
Epoch [27/50], Iter [400/418], Loss: total=2.372, reg=14.009,
containing_obj=8.220, no_obj=2.051, cls=4.183
Time for epoch: 372.39 seconds
Updating best test loss: 2.71612

Starting epoch 28 / 50

Learning Rate for this epoch: 0.001

Epoch [28/50], Iter [50/418], Loss: total=2.319, reg=13.811,
containing_obj=7.985, no_obj=2.143, cls=3.892
Epoch [28/50], Iter [100/418], Loss: total=2.299, reg=13.628,
containing_obj=7.919, no_obj=2.105, cls=3.941
Epoch [28/50], Iter [150/418], Loss: total=2.392, reg=14.302,
containing_obj=8.163, no_obj=2.100, cls=4.141
Epoch [28/50], Iter [200/418], Loss: total=2.360, reg=14.114,
containing_obj=8.044, no_obj=2.086, cls=4.072
Epoch [28/50], Iter [250/418], Loss: total=2.374, reg=14.169,
containing_obj=8.133, no_obj=2.068, cls=4.116
Epoch [28/50], Iter [300/418], Loss: total=2.391, reg=14.297,
containing_obj=8.189, no_obj=2.069, cls=4.139
Epoch [28/50], Iter [350/418], Loss: total=2.390, reg=14.313,
containing_obj=8.147, no_obj=2.085, cls=4.129
Epoch [28/50], Iter [400/418], Loss: total=2.390, reg=14.322,
containing_obj=8.124, no_obj=2.091, cls=4.142
Time for epoch: 1161.87 seconds
Updating best test loss: 2.67412

Starting epoch 29 / 50

Learning Rate for this epoch: 0.001

Epoch [29/50], Iter [50/418], Loss: total=2.344, reg=14.084,
containing_obj=8.146, no_obj=2.037, cls=3.860
Epoch [29/50], Iter [100/418], Loss: total=2.405, reg=14.451,
containing_obj=8.273, no_obj=2.138, cls=3.994
Epoch [29/50], Iter [150/418], Loss: total=2.387, reg=14.211,
containing_obj=8.208, no_obj=2.149, cls=4.079
Epoch [29/50], Iter [200/418], Loss: total=2.323, reg=13.836,
containing_obj=8.101, no_obj=2.109, cls=3.826
Epoch [29/50], Iter [250/418], Loss: total=2.316, reg=13.802,
containing_obj=8.090, no_obj=2.091, cls=3.812
Epoch [29/50], Iter [300/418], Loss: total=2.324, reg=13.840,
containing_obj=8.129, no_obj=2.088, cls=3.836
Epoch [29/50], Iter [350/418], Loss: total=2.309, reg=13.683,
containing_obj=8.119, no_obj=2.085, cls=3.826
Epoch [29/50], Iter [400/418], Loss: total=2.312, reg=13.715,
containing_obj=8.101, no_obj=2.091, cls=3.833
Time for epoch: 288.42 seconds

Starting epoch 30 / 50

Learning Rate for this epoch: 0.001

Epoch [30/50], Iter [50/418], Loss: total=2.340, reg=13.401,
containing_obj=8.363, no_obj=2.225, cls=4.087
Epoch [30/50], Iter [100/418], Loss: total=2.226, reg=12.917,
containing_obj=7.844, no_obj=2.233, cls=3.715
Epoch [30/50], Iter [150/418], Loss: total=2.266, reg=13.274,
containing_obj=7.908, no_obj=2.186, cls=3.829
Epoch [30/50], Iter [200/418], Loss: total=2.281, reg=13.372,
containing_obj=7.939, no_obj=2.203, cls=3.853
Epoch [30/50], Iter [250/418], Loss: total=2.273, reg=13.364,
containing_obj=7.917, no_obj=2.189, cls=3.801
Epoch [30/50], Iter [300/418], Loss: total=2.280, reg=13.474,
containing_obj=7.947, no_obj=2.164, cls=3.781
Epoch [30/50], Iter [350/418], Loss: total=2.278, reg=13.495,
containing_obj=7.978, no_obj=2.157, cls=3.702
Epoch [30/50], Iter [400/418], Loss: total=2.281, reg=13.495,
containing_obj=8.033, no_obj=2.145, cls=3.698
Time for epoch: 1560.85 seconds

---Evaluate model on test samples---

100%| | 4950/4950 [02:14<00:00, 36.71it/s]

---class aeroplane ap 0.48394455471620573---

---class bicycle ap 0.5945576143043148---

---class bird ap 0.4636881969138623---

---class boat ap 0.2795500526564305---

```

---class bottle ap 0.1453420488813854---
---class bus ap 0.620578878750474---
---class car ap 0.6370162605931345---
---class cat ap 0.6435214321982385---
---class chair ap 0.24541515486297721---
---class cow ap 0.44419361077720676---
---class diningtable ap 0.2782678106991446---
---class dog ap 0.622111566329097---
---class horse ap 0.6207886570771708---
---class motorbike ap 0.5645605424490072---
---class person ap 0.5082784752016805---
---class pottedplant ap 0.12610590815978656---
---class sheep ap 0.46039434809787494---
---class sofa ap 0.4426041826180284---
---class train ap 0.6236789424777944---
---class tvmonitor ap 0.4722516876785181---
---map 0.4638424962721167---
29 [np.float64(0.48394455471620573), np.float64(0.5945576143043148),
np.float64(0.4636881969138623), np.float64(0.2795500526564305),
np.float64(0.1453420488813854), np.float64(0.620578878750474),
np.float64(0.6370162605931345), np.float64(0.6435214321982385),
np.float64(0.24541515486297721), np.float64(0.44419361077720676),
np.float64(0.2782678106991446), np.float64(0.622111566329097),
np.float64(0.6207886570771708), np.float64(0.5645605424490072),
np.float64(0.5082784752016805), np.float64(0.12610590815978656),
np.float64(0.46039434809787494), np.float64(0.4426041826180284),
np.float64(0.6236789424777944), np.float64(0.4722516876785181)]
Updating best test loss: 2.66434

```

Starting epoch 31 / 50

Learning Rate for this epoch: 0.0001

```

Epoch [31/50], Iter [50/418], Loss: total=2.175, reg=12.700,
containing_obj=8.034, no_obj=2.165, cls=3.204
Epoch [31/50], Iter [100/418], Loss: total=2.152, reg=12.524,
containing_obj=7.737, no_obj=2.144, cls=3.420
Epoch [31/50], Iter [150/418], Loss: total=2.223, reg=13.045,
containing_obj=7.982, no_obj=2.124, cls=3.522
Epoch [31/50], Iter [200/418], Loss: total=2.230, reg=13.084,
containing_obj=8.011, no_obj=2.136, cls=3.532
Epoch [31/50], Iter [250/418], Loss: total=2.220, reg=13.016,
containing_obj=7.982, no_obj=2.141, cls=3.501
Epoch [31/50], Iter [300/418], Loss: total=2.224, reg=13.071,
containing_obj=7.998, no_obj=2.135, cls=3.482
Epoch [31/50], Iter [350/418], Loss: total=2.232, reg=13.097,
containing_obj=8.040, no_obj=2.140, cls=3.503
Epoch [31/50], Iter [400/418], Loss: total=2.218, reg=13.043,
containing_obj=7.945, no_obj=2.138, cls=3.490

```

Time for epoch: 302.36 seconds
Updating best test loss: 2.63595

Starting epoch 32 / 50

Learning Rate for this epoch: 0.0001

Epoch [32/50], Iter [50/418], Loss: total=2.074, reg=12.337,
containing_obj=7.392, no_obj=2.085, cls=3.073
Epoch [32/50], Iter [100/418], Loss: total=2.127, reg=12.879,
containing_obj=7.476, no_obj=2.124, cls=3.050
Epoch [32/50], Iter [150/418], Loss: total=2.103, reg=12.603,
containing_obj=7.524, no_obj=2.152, cls=2.955
Epoch [32/50], Iter [200/418], Loss: total=2.151, reg=12.861,
containing_obj=7.683, no_obj=2.140, cls=3.129
Epoch [32/50], Iter [250/418], Loss: total=2.197, reg=13.073,
containing_obj=7.895, no_obj=2.130, cls=3.264
Epoch [32/50], Iter [300/418], Loss: total=2.175, reg=12.923,
containing_obj=7.795, no_obj=2.148, cls=3.231
Epoch [32/50], Iter [350/418], Loss: total=2.192, reg=13.070,
containing_obj=7.797, no_obj=2.158, cls=3.276
Epoch [32/50], Iter [400/418], Loss: total=2.191, reg=13.023,
containing_obj=7.836, no_obj=2.170, cls=3.267

Time for epoch: 1208.41 seconds

Updating best test loss: 2.62305

Starting epoch 33 / 50

Learning Rate for this epoch: 0.0001

Epoch [33/50], Iter [50/418], Loss: total=2.159, reg=12.698,
containing_obj=7.677, no_obj=2.152, cls=3.384
Epoch [33/50], Iter [100/418], Loss: total=2.188, reg=12.929,
containing_obj=7.699, no_obj=2.168, cls=3.465
Epoch [33/50], Iter [150/418], Loss: total=2.163, reg=12.791,
containing_obj=7.703, no_obj=2.183, cls=3.273
Epoch [33/50], Iter [200/418], Loss: total=2.152, reg=12.842,
containing_obj=7.612, no_obj=2.187, cls=3.181
Epoch [33/50], Iter [250/418], Loss: total=2.146, reg=12.726,
containing_obj=7.592, no_obj=2.175, cls=3.259
Epoch [33/50], Iter [300/418], Loss: total=2.162, reg=12.816,
containing_obj=7.686, no_obj=2.163, cls=3.283
Epoch [33/50], Iter [350/418], Loss: total=2.166, reg=12.755,
containing_obj=7.737, no_obj=2.172, cls=3.326
Epoch [33/50], Iter [400/418], Loss: total=2.168, reg=12.748,
containing_obj=7.745, no_obj=2.171, cls=3.348

Time for epoch: 230.05 seconds

Updating best test loss: 2.60314

Starting epoch 34 / 50

Learning Rate for this epoch: 0.0001

Epoch [34/50], Iter [50/418], Loss: total=2.201, reg=12.580,
containing_obj=8.267, no_obj=2.239, cls=3.325
Epoch [34/50], Iter [100/418], Loss: total=2.238, reg=13.140,
containing_obj=8.129, no_obj=2.181, cls=3.405
Epoch [34/50], Iter [150/418], Loss: total=2.256, reg=13.260,
containing_obj=8.089, no_obj=2.182, cls=3.535
Epoch [34/50], Iter [200/418], Loss: total=2.203, reg=12.852,
containing_obj=7.946, no_obj=2.196, cls=3.439
Epoch [34/50], Iter [250/418], Loss: total=2.178, reg=12.799,
containing_obj=7.866, no_obj=2.212, cls=3.264
Epoch [34/50], Iter [300/418], Loss: total=2.145, reg=12.651,
containing_obj=7.686, no_obj=2.224, cls=3.174
Epoch [34/50], Iter [350/418], Loss: total=2.124, reg=12.538,
containing_obj=7.622, no_obj=2.218, cls=3.105
Epoch [34/50], Iter [400/418], Loss: total=2.132, reg=12.636,
containing_obj=7.625, no_obj=2.216, cls=3.110
Time for epoch: 191.63 seconds

Starting epoch 35 / 50

Learning Rate for this epoch: 0.0001

Epoch [35/50], Iter [50/418], Loss: total=2.106, reg=12.576,
containing_obj=7.480, no_obj=2.247, cls=2.968
Epoch [35/50], Iter [100/418], Loss: total=2.077, reg=12.129,
containing_obj=7.399, no_obj=2.235, cls=3.156
Epoch [35/50], Iter [150/418], Loss: total=2.124, reg=12.466,
containing_obj=7.580, no_obj=2.239, cls=3.199
Epoch [35/50], Iter [200/418], Loss: total=2.124, reg=12.451,
containing_obj=7.650, no_obj=2.225, cls=3.161
Epoch [35/50], Iter [250/418], Loss: total=2.134, reg=12.608,
containing_obj=7.677, no_obj=2.228, cls=3.090
Epoch [35/50], Iter [300/418], Loss: total=2.149, reg=12.757,
containing_obj=7.703, no_obj=2.223, cls=3.101
Epoch [35/50], Iter [350/418], Loss: total=2.152, reg=12.742,
containing_obj=7.736, no_obj=2.220, cls=3.124
Epoch [35/50], Iter [400/418], Loss: total=2.152, reg=12.751,
containing_obj=7.737, no_obj=2.210, cls=3.122
Time for epoch: 632.24 seconds

---Evaluate model on test samples---

100%| | 4950/4950 [01:24<00:00, 58.74it/s]

---class aeroplane ap 0.47303652735450674---
---class bicycle ap 0.6133461280298301---
---class bird ap 0.4801669834987738---
---class boat ap 0.2743416206493816---
---class bottle ap 0.16702195790618674---

```

---class bus ap 0.6126386254485294---
---class car ap 0.6302306550790444---
---class cat ap 0.6830346393135203---
---class chair ap 0.2835546931749364---
---class cow ap 0.5077472492749946---
---class diningtable ap 0.29202458184745916---
---class dog ap 0.6468169151454483---
---class horse ap 0.6493478968179793---
---class motorbike ap 0.5719525819679141---
---class person ap 0.49719871434029106---
---class pottedplant ap 0.15746010806900873---
---class sheep ap 0.4852487336947293---
---class sofa ap 0.4908731097682495---
---class train ap 0.6433778024582485---
---class tvmonitor ap 0.4834056444311131---
---map 0.48214125841350725---
34 [np.float64(0.47303652735450674), np.float64(0.6133461280298301),
np.float64(0.4801669834987738), np.float64(0.2743416206493816),
np.float64(0.16702195790618674), np.float64(0.6126386254485294),
np.float64(0.6302306550790444), np.float64(0.6830346393135203),
np.float64(0.2835546931749364), np.float64(0.5077472492749946),
np.float64(0.29202458184745916), np.float64(0.6468169151454483),
np.float64(0.6493478968179793), np.float64(0.5719525819679141),
np.float64(0.49719871434029106), np.float64(0.15746010806900873),
np.float64(0.4852487336947293), np.float64(0.4908731097682495),
np.float64(0.6433778024582485), np.float64(0.4834056444311131)]
Updating best test loss: 2.59084

```

Starting epoch 36 / 50

Learning Rate for this epoch: 0.0001

```

Epoch [36/50], Iter [50/418], Loss: total=2.038, reg=11.844,
containing_obj=7.258, no_obj=2.278, cls=3.070
Epoch [36/50], Iter [100/418], Loss: total=1.985, reg=11.389,
containing_obj=7.174, no_obj=2.267, cls=2.991
Epoch [36/50], Iter [150/418], Loss: total=2.068, reg=12.020,
containing_obj=7.530, no_obj=2.197, cls=3.070
Epoch [36/50], Iter [200/418], Loss: total=2.062, reg=11.910,
containing_obj=7.518, no_obj=2.220, cls=3.094
Epoch [36/50], Iter [250/418], Loss: total=2.089, reg=12.127,
containing_obj=7.548, no_obj=2.199, cls=3.200
Epoch [36/50], Iter [300/418], Loss: total=2.088, reg=12.167,
containing_obj=7.545, no_obj=2.210, cls=3.134
Epoch [36/50], Iter [350/418], Loss: total=2.115, reg=12.362,
containing_obj=7.661, no_obj=2.202, cls=3.158
Epoch [36/50], Iter [400/418], Loss: total=2.109, reg=12.321,
containing_obj=7.642, no_obj=2.209, cls=3.134
Time for epoch: 191.96 seconds

```

Updating best test loss: 2.58743

Starting epoch 37 / 50

Learning Rate for this epoch: 0.0001

Epoch [37/50], Iter [50/418], Loss: total=1.942, reg=11.446,
containing_obj=7.063, no_obj=2.264, cls=2.531
Epoch [37/50], Iter [100/418], Loss: total=2.040, reg=11.894,
containing_obj=7.500, no_obj=2.257, cls=2.830
Epoch [37/50], Iter [150/418], Loss: total=2.115, reg=12.355,
containing_obj=7.748, no_obj=2.217, cls=3.061
Epoch [37/50], Iter [200/418], Loss: total=2.121, reg=12.399,
containing_obj=7.797, no_obj=2.216, cls=3.035
Epoch [37/50], Iter [250/418], Loss: total=2.095, reg=12.259,
containing_obj=7.689, no_obj=2.220, cls=2.971
Epoch [37/50], Iter [300/418], Loss: total=2.133, reg=12.529,
containing_obj=7.812, no_obj=2.203, cls=3.053
Epoch [37/50], Iter [350/418], Loss: total=2.114, reg=12.341,
containing_obj=7.769, no_obj=2.198, cls=3.054
Epoch [37/50], Iter [400/418], Loss: total=2.121, reg=12.425,
containing_obj=7.762, no_obj=2.205, cls=3.057
Time for epoch: 191.28 seconds
Updating best test loss: 2.58546

Starting epoch 38 / 50

Learning Rate for this epoch: 0.0001

Epoch [38/50], Iter [50/418], Loss: total=2.106, reg=11.900,
containing_obj=7.720, no_obj=2.233, cls=3.423
Epoch [38/50], Iter [100/418], Loss: total=2.125, reg=12.193,
containing_obj=7.803, no_obj=2.213, cls=3.290
Epoch [38/50], Iter [150/418], Loss: total=2.106, reg=12.081,
containing_obj=7.784, no_obj=2.231, cls=3.171
Epoch [38/50], Iter [200/418], Loss: total=2.101, reg=12.176,
containing_obj=7.740, no_obj=2.212, cls=3.084
Epoch [38/50], Iter [250/418], Loss: total=2.120, reg=12.446,
containing_obj=7.731, no_obj=2.212, cls=3.050
Epoch [38/50], Iter [300/418], Loss: total=2.131, reg=12.490,
containing_obj=7.773, no_obj=2.227, cls=3.083
Epoch [38/50], Iter [350/418], Loss: total=2.121, reg=12.404,
containing_obj=7.741, no_obj=2.243, cls=3.062
Epoch [38/50], Iter [400/418], Loss: total=2.113, reg=12.296,
containing_obj=7.729, no_obj=2.241, cls=3.089
Time for epoch: 191.66 seconds

Starting epoch 39 / 50

Learning Rate for this epoch: 0.0001

Epoch [39/50], Iter [50/418], Loss: total=1.998, reg=11.620,
containing_obj=7.230, no_obj=2.340, cls=2.788
Epoch [39/50], Iter [100/418], Loss: total=2.081, reg=12.177,
containing_obj=7.658, no_obj=2.247, cls=2.887
Epoch [39/50], Iter [150/418], Loss: total=2.065, reg=12.054,
containing_obj=7.570, no_obj=2.242, cls=2.917
Epoch [39/50], Iter [200/418], Loss: total=2.089, reg=12.241,
containing_obj=7.614, no_obj=2.282, cls=2.925
Epoch [39/50], Iter [250/418], Loss: total=2.099, reg=12.367,
containing_obj=7.595, no_obj=2.255, cls=2.970
Epoch [39/50], Iter [300/418], Loss: total=2.090, reg=12.268,
containing_obj=7.576, no_obj=2.261, cls=2.977
Epoch [39/50], Iter [350/418], Loss: total=2.100, reg=12.369,
containing_obj=7.603, no_obj=2.279, cls=2.943
Epoch [39/50], Iter [400/418], Loss: total=2.113, reg=12.488,
containing_obj=7.605, no_obj=2.269, cls=2.990
Time for epoch: 191.03 seconds
Updating best test loss: 2.57873

Starting epoch 40 / 50

Learning Rate for this epoch: 0.0001

Epoch [40/50], Iter [50/418], Loss: total=2.062, reg=12.382,
containing_obj=7.206, no_obj=2.222, cls=2.939
Epoch [40/50], Iter [100/418], Loss: total=2.113, reg=12.476,
containing_obj=7.622, no_obj=2.162, cls=3.097
Epoch [40/50], Iter [150/418], Loss: total=2.138, reg=12.561,
containing_obj=7.797, no_obj=2.166, cls=3.134
Epoch [40/50], Iter [200/418], Loss: total=2.138, reg=12.616,
containing_obj=7.824, no_obj=2.170, cls=3.051
Epoch [40/50], Iter [250/418], Loss: total=2.122, reg=12.510,
containing_obj=7.789, no_obj=2.200, cls=2.965
Epoch [40/50], Iter [300/418], Loss: total=2.126, reg=12.460,
containing_obj=7.879, no_obj=2.185, cls=2.985
Epoch [40/50], Iter [350/418], Loss: total=2.112, reg=12.340,
containing_obj=7.840, no_obj=2.197, cls=2.969
Epoch [40/50], Iter [400/418], Loss: total=2.092, reg=12.252,
containing_obj=7.688, no_obj=2.211, cls=2.950
Time for epoch: 190.90 seconds

---Evaluate model on test samples---

100%| | 4950/4950 [01:22<00:00, 60.13it/s]

---class aeroplane ap 0.4619237624804502---
---class bicycle ap 0.6116033793929083---
---class bird ap 0.4908453789513034---
---class boat ap 0.2845099242330523---
---class bottle ap 0.1641490300602611---
---class bus ap 0.6124302518182851---

```

---class car ap 0.6349691594951739---
---class cat ap 0.6993044814360355---
---class chair ap 0.2834558974393311---
---class cow ap 0.489023568064647---
---class diningtable ap 0.32151221811772634---
---class dog ap 0.6712368210174051---
---class horse ap 0.624054580830343---
---class motorbike ap 0.5440455514089293---
---class person ap 0.5150582115856392---
---class pottedplant ap 0.13635086205794308---
---class sheep ap 0.4487596848322001---
---class sofa ap 0.5029017862271692---
---class train ap 0.6540783722146637---
---class tvmonitor ap 0.468280821754909---
---map 0.48092468717091885---
39 [np.float64(0.4619237624804502), np.float64(0.6116033793929083),
np.float64(0.4908453789513034), np.float64(0.2845099242330523),
np.float64(0.1641490300602611), np.float64(0.6124302518182851),
np.float64(0.6349691594951739), np.float64(0.6993044814360355),
np.float64(0.2834558974393311), np.float64(0.489023568064647),
np.float64(0.32151221811772634), np.float64(0.6712368210174051),
np.float64(0.624054580830343), np.float64(0.5440455514089293),
np.float64(0.5150582115856392), np.float64(0.13635086205794308),
np.float64(0.4487596848322001), np.float64(0.5029017862271692),
np.float64(0.6540783722146637), np.float64(0.468280821754909)]
Updating best test loss: 2.57127

```

```

Starting epoch 41 / 50
Learning Rate for this epoch: 1e-05
Epoch [41/50], Iter [50/418], Loss: total=2.041, reg=11.626,
containing_obj=7.661, no_obj=2.294, cls=2.908
Epoch [41/50], Iter [100/418], Loss: total=2.034, reg=11.739,
containing_obj=7.568, no_obj=2.274, cls=2.833
Epoch [41/50], Iter [150/418], Loss: total=2.010, reg=11.582,
containing_obj=7.467, no_obj=2.278, cls=2.796
Epoch [41/50], Iter [200/418], Loss: total=1.997, reg=11.566,
containing_obj=7.319, no_obj=2.274, cls=2.804
Epoch [41/50], Iter [250/418], Loss: total=2.056, reg=12.000,
containing_obj=7.522, no_obj=2.258, cls=2.891
Epoch [41/50], Iter [300/418], Loss: total=2.062, reg=12.020,
containing_obj=7.549, no_obj=2.266, cls=2.909
Epoch [41/50], Iter [350/418], Loss: total=2.056, reg=12.023,
containing_obj=7.538, no_obj=2.247, cls=2.860
Epoch [41/50], Iter [400/418], Loss: total=2.058, reg=12.044,
containing_obj=7.562, no_obj=2.251, cls=2.842
Time for epoch: 2076.63 seconds

```

Starting epoch 42 / 50

Learning Rate for this epoch: 1e-05

Epoch [42/50], Iter [50/418], Loss: total=2.097, reg=12.469,
containing_obj=7.677, no_obj=2.189, cls=2.824
Epoch [42/50], Iter [100/418], Loss: total=2.146, reg=12.590,
containing_obj=7.997, no_obj=2.191, cls=2.978
Epoch [42/50], Iter [150/418], Loss: total=2.145, reg=12.535,
containing_obj=8.026, no_obj=2.204, cls=2.969
Epoch [42/50], Iter [200/418], Loss: total=2.076, reg=12.164,
containing_obj=7.635, no_obj=2.224, cls=2.888
Epoch [42/50], Iter [250/418], Loss: total=2.063, reg=12.053,
containing_obj=7.557, no_obj=2.238, cls=2.914
Epoch [42/50], Iter [300/418], Loss: total=2.085, reg=12.165,
containing_obj=7.674, no_obj=2.232, cls=2.953
Epoch [42/50], Iter [350/418], Loss: total=2.091, reg=12.197,
containing_obj=7.679, no_obj=2.233, cls=2.985
Epoch [42/50], Iter [400/418], Loss: total=2.086, reg=12.165,
containing_obj=7.648, no_obj=2.238, cls=2.980
Time for epoch: 302.43 seconds

Starting epoch 43 / 50

Learning Rate for this epoch: 1e-05

Epoch [43/50], Iter [50/418], Loss: total=1.912, reg=11.196,
containing_obj=7.081, no_obj=2.275, cls=2.388
Epoch [43/50], Iter [100/418], Loss: total=1.970, reg=11.406,
containing_obj=7.277, no_obj=2.267, cls=2.689
Epoch [43/50], Iter [150/418], Loss: total=2.021, reg=11.756,
containing_obj=7.480, no_obj=2.297, cls=2.726
Epoch [43/50], Iter [200/418], Loss: total=2.039, reg=11.761,
containing_obj=7.539, no_obj=2.310, cls=2.861
Epoch [43/50], Iter [250/418], Loss: total=2.040, reg=11.783,
containing_obj=7.531, no_obj=2.306, cls=2.855
Epoch [43/50], Iter [300/418], Loss: total=2.061, reg=11.984,
containing_obj=7.627, no_obj=2.277, cls=2.849
Epoch [43/50], Iter [350/418], Loss: total=2.057, reg=11.984,
containing_obj=7.579, no_obj=2.266, cls=2.853
Epoch [43/50], Iter [400/418], Loss: total=2.069, reg=12.072,
containing_obj=7.627, no_obj=2.248, cls=2.886
Time for epoch: 306.33 seconds
Updating best test loss: 2.56956

Starting epoch 44 / 50

Learning Rate for this epoch: 1e-05

Epoch [44/50], Iter [50/418], Loss: total=2.159, reg=12.939,
containing_obj=7.920, no_obj=2.156, cls=2.896

```
Epoch [44/50], Iter [100/418], Loss: total=2.158, reg=12.801,
containing_obj=7.745, no_obj=2.180, cls=3.168
Epoch [44/50], Iter [150/418], Loss: total=2.110, reg=12.446,
containing_obj=7.608, no_obj=2.251, cls=3.021
Epoch [44/50], Iter [200/418], Loss: total=2.129, reg=12.516,
containing_obj=7.681, no_obj=2.233, cls=3.120
Epoch [44/50], Iter [250/418], Loss: total=2.110, reg=12.350,
containing_obj=7.642, no_obj=2.221, cls=3.104
Epoch [44/50], Iter [300/418], Loss: total=2.100, reg=12.299,
containing_obj=7.646, no_obj=2.230, cls=3.028
Epoch [44/50], Iter [350/418], Loss: total=2.085, reg=12.242,
containing_obj=7.558, no_obj=2.224, cls=2.995
Epoch [44/50], Iter [400/418], Loss: total=2.074, reg=12.163,
containing_obj=7.545, no_obj=2.218, cls=2.958
Time for epoch: 307.44 seconds
```

Starting epoch 45 / 50

Learning Rate for this epoch: 1e-05

```
Epoch [45/50], Iter [50/418], Loss: total=1.996, reg=11.446,
containing_obj=7.369, no_obj=2.299, cls=2.836
Epoch [45/50], Iter [100/418], Loss: total=2.069, reg=12.111,
containing_obj=7.592, no_obj=2.265, cls=2.863
Epoch [45/50], Iter [150/418], Loss: total=2.080, reg=12.274,
containing_obj=7.610, no_obj=2.227, cls=2.846
Epoch [45/50], Iter [200/418], Loss: total=2.121, reg=12.577,
containing_obj=7.778, no_obj=2.221, cls=2.880
Epoch [45/50], Iter [250/418], Loss: total=2.095, reg=12.334,
containing_obj=7.729, no_obj=2.231, cls=2.847
Epoch [45/50], Iter [300/418], Loss: total=2.082, reg=12.216,
containing_obj=7.695, no_obj=2.234, cls=2.834
Epoch [45/50], Iter [350/418], Loss: total=2.063, reg=12.099,
containing_obj=7.590, no_obj=2.246, cls=2.823
Epoch [45/50], Iter [400/418], Loss: total=2.073, reg=12.163,
containing_obj=7.623, no_obj=2.248, cls=2.847
Time for epoch: 311.00 seconds
```

---Evaluate model on test samples---

100%| | 4950/4950 [02:19<00:00, 35.52it/s]

```
---class aeroplane ap 0.4603398514139901---
---class bicycle ap 0.6216907883112832---
---class bird ap 0.5187309863355247---
---class boat ap 0.27628111397501887---
---class bottle ap 0.16962369624688434---
---class bus ap 0.597441727275696---
---class car ap 0.632757968890833---
---class cat ap 0.7014925632720523---
---class chair ap 0.2969930475001427---
```

```

---class cow ap 0.5369707618010302---
---class diningtable ap 0.28012050010165074---
---class dog ap 0.6798775433243365---
---class horse ap 0.6610184233259275---
---class motorbike ap 0.5562772974584052---
---class person ap 0.5206675674827165---
---class pottedplant ap 0.14711667850738136---
---class sheep ap 0.5408383019623909---
---class sofa ap 0.478500689319301---
---class train ap 0.6375051750980205---
---class tvmonitor ap 0.5169098866839723---
---map 0.4915577284143279---
44 [np.float64(0.4603398514139901), np.float64(0.6216907883112832),
np.float64(0.5187309863355247), np.float64(0.27628111397501887),
np.float64(0.16962369624688434), np.float64(0.597441727275696),
np.float64(0.632757968890833), np.float64(0.7014925632720523),
np.float64(0.2969930475001427), np.float64(0.5369707618010302),
np.float64(0.28012050010165074), np.float64(0.6798775433243365),
np.float64(0.6610184233259275), np.float64(0.5562772974584052),
np.float64(0.5206675674827165), np.float64(0.14711667850738136),
np.float64(0.5408383019623909), np.float64(0.478500689319301),
np.float64(0.6375051750980205), np.float64(0.5169098866839723)]
Updating best test loss: 2.56290

```

Starting epoch 46 / 50

Learning Rate for this epoch: 1e-05

```

Epoch [46/50], Iter [50/418], Loss: total=2.099, reg=12.420,
containing_obj=7.654, no_obj=2.119, cls=2.994
Epoch [46/50], Iter [100/418], Loss: total=2.045, reg=12.070,
containing_obj=7.490, no_obj=2.175, cls=2.804
Epoch [46/50], Iter [150/418], Loss: total=2.057, reg=12.004,
containing_obj=7.598, no_obj=2.217, cls=2.869
Epoch [46/50], Iter [200/418], Loss: total=2.068, reg=12.037,
containing_obj=7.629, no_obj=2.220, cls=2.934
Epoch [46/50], Iter [250/418], Loss: total=2.071, reg=12.067,
containing_obj=7.606, no_obj=2.226, cls=2.947
Epoch [46/50], Iter [300/418], Loss: total=2.047, reg=11.911,
containing_obj=7.497, no_obj=2.241, cls=2.918
Epoch [46/50], Iter [350/418], Loss: total=2.048, reg=11.921,
containing_obj=7.532, no_obj=2.220, cls=2.901
Epoch [46/50], Iter [400/418], Loss: total=2.047, reg=11.922,
containing_obj=7.556, no_obj=2.211, cls=2.873
Time for epoch: 308.76 seconds

```

Starting epoch 47 / 50

Learning Rate for this epoch: 1e-05

Epoch [47/50], Iter [50/418], Loss: total=2.230, reg=13.465,
containing_obj=8.098, no_obj=2.198, cls=3.003
Epoch [47/50], Iter [100/418], Loss: total=2.098, reg=12.491,
containing_obj=7.646, no_obj=2.258, cls=2.780
Epoch [47/50], Iter [150/418], Loss: total=2.041, reg=12.033,
containing_obj=7.414, no_obj=2.284, cls=2.765
Epoch [47/50], Iter [200/418], Loss: total=2.057, reg=12.197,
containing_obj=7.379, no_obj=2.265, cls=2.846
Epoch [47/50], Iter [250/418], Loss: total=2.070, reg=12.272,
containing_obj=7.452, no_obj=2.247, cls=2.864
Epoch [47/50], Iter [300/418], Loss: total=2.065, reg=12.207,
containing_obj=7.489, no_obj=2.244, cls=2.845
Epoch [47/50], Iter [350/418], Loss: total=2.060, reg=12.140,
containing_obj=7.520, no_obj=2.228, cls=2.835
Epoch [47/50], Iter [400/418], Loss: total=2.062, reg=12.139,
containing_obj=7.521, no_obj=2.226, cls=2.853
Time for epoch: 309.53 seconds

Starting epoch 48 / 50

Learning Rate for this epoch: 1e-05

Epoch [48/50], Iter [50/418], Loss: total=2.093, reg=12.314,
containing_obj=7.814, no_obj=2.108, cls=2.881
Epoch [48/50], Iter [100/418], Loss: total=2.076, reg=12.235,
containing_obj=7.582, no_obj=2.157, cls=2.942
Epoch [48/50], Iter [150/418], Loss: total=2.065, reg=12.175,
containing_obj=7.461, no_obj=2.187, cls=2.959
Epoch [48/50], Iter [200/418], Loss: total=2.069, reg=12.156,
containing_obj=7.523, no_obj=2.213, cls=2.931
Epoch [48/50], Iter [250/418], Loss: total=2.086, reg=12.347,
containing_obj=7.559, no_obj=2.220, cls=2.901
Epoch [48/50], Iter [300/418], Loss: total=2.075, reg=12.259,
containing_obj=7.558, no_obj=2.220, cls=2.861
Epoch [48/50], Iter [350/418], Loss: total=2.062, reg=12.144,
containing_obj=7.525, no_obj=2.226, cls=2.844
Epoch [48/50], Iter [400/418], Loss: total=2.070, reg=12.219,
containing_obj=7.560, no_obj=2.230, cls=2.837
Time for epoch: 311.34 seconds

Starting epoch 49 / 50

Learning Rate for this epoch: 1e-05

Epoch [49/50], Iter [50/418], Loss: total=2.172, reg=12.928,
containing_obj=8.028, no_obj=2.135, cls=2.977
Epoch [49/50], Iter [100/418], Loss: total=2.135, reg=12.428,
containing_obj=7.900, no_obj=2.180, cls=3.106
Epoch [49/50], Iter [150/418], Loss: total=2.109, reg=12.356,
containing_obj=7.777, no_obj=2.192, cls=2.980

Epoch [49/50], Iter [200/418], Loss: total=2.112, reg=12.357,
containing_obj=7.810, no_obj=2.178, cls=2.995
Epoch [49/50], Iter [250/418], Loss: total=2.109, reg=12.357,
containing_obj=7.795, no_obj=2.192, cls=2.969
Epoch [49/50], Iter [300/418], Loss: total=2.084, reg=12.184,
containing_obj=7.697, no_obj=2.212, cls=2.909
Epoch [49/50], Iter [350/418], Loss: total=2.076, reg=12.093,
containing_obj=7.684, no_obj=2.226, cls=2.913
Epoch [49/50], Iter [400/418], Loss: total=2.072, reg=12.046,
containing_obj=7.645, no_obj=2.234, cls=2.934
Time for epoch: 188.01 seconds

Starting epoch 50 / 50

Learning Rate for this epoch: 1e-05

Epoch [50/50], Iter [50/418], Loss: total=2.056, reg=12.136,
containing_obj=7.524, no_obj=2.189, cls=2.827
Epoch [50/50], Iter [100/418], Loss: total=2.114, reg=12.586,
containing_obj=7.671, no_obj=2.213, cls=2.898
Epoch [50/50], Iter [150/418], Loss: total=2.068, reg=12.242,
containing_obj=7.654, no_obj=2.205, cls=2.716
Epoch [50/50], Iter [200/418], Loss: total=2.056, reg=12.134,
containing_obj=7.604, no_obj=2.206, cls=2.729
Epoch [50/50], Iter [250/418], Loss: total=2.045, reg=12.040,
containing_obj=7.525, no_obj=2.220, cls=2.757
Epoch [50/50], Iter [300/418], Loss: total=2.034, reg=11.951,
containing_obj=7.520, no_obj=2.220, cls=2.721
Epoch [50/50], Iter [350/418], Loss: total=2.036, reg=11.938,
containing_obj=7.521, no_obj=2.211, cls=2.757
Epoch [50/50], Iter [400/418], Loss: total=2.060, reg=12.076,
containing_obj=7.594, no_obj=2.199, cls=2.848
Time for epoch: 187.76 seconds

---Evaluate model on test samples---

100%| | 4950/4950 [01:20<00:00, 61.24it/s]

---class aeroplane ap 0.46927644599075197---
---class bicycle ap 0.6089012954218461---
---class bird ap 0.48156144379732746---
---class boat ap 0.28344146221136723---
---class bottle ap 0.16111567200211296---
---class bus ap 0.6156163463559517---
---class car ap 0.6330781363091106---
---class cat ap 0.7073445652220068---
---class chair ap 0.27550388240078194---
---class cow ap 0.49205767436548087---
---class diningtable ap 0.32167899716874554---
---class dog ap 0.6542053492054896---
---class horse ap 0.6169559817382446---

```

---class motorbike ap 0.5670876772938439---
---class person ap 0.5084109799689435---
---class pottedplant ap 0.14515886773403983---
---class sheep ap 0.4550508804767739---
---class sofa ap 0.509445432154262---
---class train ap 0.6745990045661254---
---class tvmonitor ap 0.4813248467764101---
---map 0.4830907470579808---
49 [np.float64(0.46927644599075197), np.float64(0.6089012954218461),
np.float64(0.48156144379732746), np.float64(0.28344146221136723),
np.float64(0.16111567200211296), np.float64(0.6156163463559517),
np.float64(0.6330781363091106), np.float64(0.7073445652220068),
np.float64(0.27550388240078194), np.float64(0.49205767436548087),
np.float64(0.32167899716874554), np.float64(0.6542053492054896),
np.float64(0.6169559817382446), np.float64(0.5670876772938439),
np.float64(0.5084109799689435), np.float64(0.14515886773403983),
np.float64(0.4550508804767739), np.float64(0.509445432154262),
np.float64(0.6745990045661254), np.float64(0.4813248467764101)]

```

1 View example predictions

```

[21]: net.eval()

# select random image from test set
image_name = random.choice(test_dataset.fnames)
image = cv2.imread(os.path.join(file_root_test, image_name))
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

print('predicting...')
result = predict_image(net, image_name, root_img_directory=file_root_test)
for left_up, right_bottom, class_name, _, prob in result:
    color = COLORS[VOC_CLASSES.index(class_name)]
    cv2.rectangle(image, left_up, right_bottom, color, 2)
    label = class_name + str(round(prob, 2))
    text_size, baseline = cv2.getTextSize(label, cv2.FONT_HERSHEY_SIMPLEX, 0.4, 1)
    p1 = (left_up[0], left_up[1] - text_size[1])
    cv2.rectangle(image, (p1[0] - 2 // 2, p1[1] - 2 - baseline), (p1[0] +
text_size[0], p1[1] + text_size[1]),
                color, -1)
    cv2.putText(image, label, (p1[0], p1[1] + baseline), cv2.
FONT_HERSHEY_SIMPLEX, 0.4, (255, 255, 255), 1, 8)

plt.figure(figsize = (15,15))
plt.imshow(image)

```

predicting...

[21]: <matplotlib.image.AxesImage at 0x7b220539e750>



1.1 Evaluate on Test

To evaluate detection results we use mAP (mean of average precision over each class)

```
[17]: # All epochs
# Load the best model checkpoint
best_model_path = 'checkpoints_152/best_detector.pth'
if os.path.exists(best_model_path):
    print(f"Loading best model from {best_model_path}")
    net.load_state_dict(torch.load(best_model_path, map_location=device))
    net.eval()
else:
    print(f"Best model checkpoint {best_model_path} not found.")

test_aps = evaluate(net, test_dataset_file=annotation_file_test,
    img_root=file_root_test)
```

Loading best model from checkpoints_152/best_detector.pth

---Evaluate model on test samples---

100%| | 4950/4950 [02:07<00:00, 38.73it/s]

---class aeroplane ap 0.5331476519028442---

---class bicycle ap 0.6573685849563118---

---class bird ap 0.5399064705682817---

---class boat ap 0.3533092851454311---

```

---class bottle ap 0.242330827600212---
---class bus ap 0.6201028150800103---
---class car ap 0.6718599565271417---
---class cat ap 0.739169933298101---
---class chair ap 0.3130854541349436---
---class cow ap 0.5315737687708765---
---class diningtable ap 0.38008041541361043---
---class dog ap 0.6783247784858413---
---class horse ap 0.6758017755415434---
---class motorbike ap 0.5548442376130216---
---class person ap 0.5570531267112093---
---class pottedplant ap 0.24055972206747822---
---class sheep ap 0.4927717471114479---
---class sofa ap 0.5122200431204742---
---class train ap 0.6279611651425476---
---class tvmonitor ap 0.47489305232009993---
---map 0.5198182405755714---

```

1.1.1 Cell added to get intermediate mAP values for students

```

[14]: network_paths = ['./checkpoints_152/detector_epoch_%d.pth' % epoch for epoch in
    ↪ [5, 10, 20, 30, 40]]+ ['./checkpoints_152/detector.pth']
for load_network_path in network_paths:
    print('Loading saved network from {}'.format(load_network_path))
    net_loaded = resnet152().to(device)
    net_loaded.load_state_dict(torch.load(load_network_path, map_location=torch.
    ↪ device(device=device)))
    evaluate(net_loaded, test_dataset_file=annotation_file_test,
    ↪ img_root=file_root_test)

```

Loading saved network from ./checkpoints_152/detector_epoch_5.pth

---Evaluate model on test samples---

100%| | 4950/4950 [02:07<00:00, 38.87it/s]

```

---class aeroplane ap 0.2604491601364827---
---class bicycle ap 0.12487180780551312---
---class bird ap 0.2196149874130022---
---class boat ap 0.0--- (no predictions for this class)
---class bottle ap 0.03575024537178055---
---class bus ap 0.12784274123043582---
---class car ap 0.4576836973900491---
---class cat ap 0.28039885163122763---
---class chair ap 0.10968765399877138---
---class cow ap 0.004098360655737705---
---class diningtable ap 0.0--- (no predictions for this class)
---class dog ap 0.21575906839694603---
---class horse ap 0.3409880243479787---
---class motorbike ap 0.030059171597633133---

```

```

---class person ap 0.30434213402465976---
---class pottedplant ap 0.01962535004347119---
---class sheep ap 0.09407173110373701---
---class sofa ap 0.0--- (no predictions for this class)
---class train ap 0.2765336494039094---
---class tvmonitor ap 0.16964681248173244---
---map 0.15357117235165338---
Loading saved network from ./checkpoints_152/detector_epoch_10.pth
---Evaluate model on test samples---

```

```

100%|      | 4950/4950 [02:06<00:00, 39.25it/s]

```

```

---class aeroplane ap 0.4458407136954164---
---class bicycle ap 0.500048589681893---
---class bird ap 0.34688681132181465---
---class boat ap 0.17033351405058156---
---class bottle ap 0.05209951524927918---
---class bus ap 0.4577080130624602---
---class car ap 0.5868443534614973---
---class cat ap 0.5520577299194005---
---class chair ap 0.20365445791205933---
---class cow ap 0.1415623816649598---
---class diningtable ap 0.0---
---class dog ap 0.42795814975515584---
---class horse ap 0.5245181640002909---
---class motorbike ap 0.44771585548698073---
---class person ap 0.4157970866147894---
---class pottedplant ap 0.09089822781173348---
---class sheep ap 0.2804502796803251---
---class sofa ap 0.22281180829292915---
---class train ap 0.49472782054408637---
---class tvmonitor ap 0.34902672996210315---
---map 0.33554701010838783---
Loading saved network from ./checkpoints_152/detector_epoch_20.pth
---Evaluate model on test samples---

```

```

100%|      | 4950/4950 [02:28<00:00, 33.43it/s]

```

```

---class aeroplane ap 0.5076640433980006---
---class bicycle ap 0.6201554491204624---
---class bird ap 0.45520289455631646---
---class boat ap 0.29059697043370275---
---class bottle ap 0.16747379960982423---
---class bus ap 0.6036170858164882---
---class car ap 0.6557080068450285---
---class cat ap 0.7094030255026985---
---class chair ap 0.27904350306261094---
---class cow ap 0.4426220270512521---
---class diningtable ap 0.26428179519547285---
---class dog ap 0.6139862092267169---

```

```

---class horse ap 0.6534792861424317---
---class motorbike ap 0.5077950689738414---
---class person ap 0.4895370551011979---
---class pottedplant ap 0.2020148850209751---
---class sheep ap 0.44572270324077967---
---class sofa ap 0.42372005424117004---
---class train ap 0.6661599262528186---
---class tvmonitor ap 0.454560325042647---
---map 0.4726372056917218---
Loading saved network from ./checkpoints_152/detector_epoch_30.pth
---Evaluate model on test samples---

100%|      | 4950/4950 [02:45<00:00, 29.88it/s]

---class aeroplane ap 0.5238143727043874---
---class bicycle ap 0.6386459435926866---
---class bird ap 0.5067897158528705---
---class boat ap 0.30944108432732886---
---class bottle ap 0.19540491763644335---
---class bus ap 0.6079055778039233---
---class car ap 0.6848427863728942---
---class cat ap 0.6947051477536839---
---class chair ap 0.30525236233613895---
---class cow ap 0.5050004667169482---
---class diningtable ap 0.403226479278777---
---class dog ap 0.6453518830784035---
---class horse ap 0.6610235420222019---
---class motorbike ap 0.5833010339932156---
---class person ap 0.5410291610021333---
---class pottedplant ap 0.16236379585593091---
---class sheep ap 0.49164144072425064---
---class sofa ap 0.4877769299921715---
---class train ap 0.6260113711265665---
---class tvmonitor ap 0.4340243361991659---
---map 0.5003776174185062---
Loading saved network from ./checkpoints_152/detector_epoch_40.pth
---Evaluate model on test samples---

100%|      | 4950/4950 [02:53<00:00, 28.53it/s]

---class aeroplane ap 0.5068753113682603---
---class bicycle ap 0.667699521222469---
---class bird ap 0.5471617346281317---
---class boat ap 0.34474114637215125---
---class bottle ap 0.24043942406400315---
---class bus ap 0.5926462408401048---
---class car ap 0.6794277249138461---
---class cat ap 0.7338018501109771---
---class chair ap 0.3120974843664938---
---class cow ap 0.5220632564938421---

```

```

---class diningtable ap 0.37652507218522324---
---class dog ap 0.6958120882544304---
---class horse ap 0.678019095206765---
---class motorbike ap 0.5696248866864488---
---class person ap 0.5578784231204488---
---class pottedplant ap 0.261545674513865---
---class sheep ap 0.5001635685039233---
---class sofa ap 0.515005983823506---
---class train ap 0.6319254859442183---
---class tvmonitor ap 0.46370275023122526---
---map 0.5198578361425168---
Loading saved network from ./checkpoints_152/detector.pth
---Evaluate model on test samples---

100%|      | 4950/4950 [02:53<00:00, 28.51it/s]

---class aeroplane ap 0.5523552713284428---
---class bicycle ap 0.6715437618493578---
---class bird ap 0.5178398036424772---
---class boat ap 0.33176230492664227---
---class bottle ap 0.22313608019351946---
---class bus ap 0.6141644174176487---
---class car ap 0.6743672221326199---
---class cat ap 0.7475535156494013---
---class chair ap 0.3266279267030193---
---class cow ap 0.5106046286155229---
---class diningtable ap 0.40626928817239905---
---class dog ap 0.6798130198274062---
---class horse ap 0.6779853181221703---
---class motorbike ap 0.5700312086673309---
---class person ap 0.5430056180652747---
---class pottedplant ap 0.24696048004648968---
---class sheep ap 0.4799145753611839---
---class sofa ap 0.5128704242213709---
---class train ap 0.6291886870740289---
---class tvmonitor ap 0.4602789861979265---
---map 0.5188136269107115---

```

```
[18]: output_submission_csv('my_solution_152.csv', test_aps)
```